

From Fragmented Findings to Design Guidelines: A Systematic Review and Synthesis of Multimodal Feedback in Eyes-Busy and Hands-Busy Interaction

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Abstract—Users increasingly interact with digital systems while visual attention, manual resources, or both are occupied by primary tasks such as driving, cooking, rehabilitation, walking, accessibility-related activities, and high-stakes precision work. In these eyes-busy and hands-busy situations, conventional visual-only or touch-dependent interfaces may be inefficient, distracting, or unsafe. This systematic literature review synthesizes 69 unique publications examining auditory, haptic, visual, and multimodal feedback under such constraints. Following a PRISMA-based review process, the literature was analyzed across six application domains: driving, cooking and domestic tasks, physical activity and rehabilitation, accessibility, walking and mobile interaction, and surgery or other high-stakes precision tasks. The synthesis focused on feedback timing, spatial location, encoding, intensity, synchronization, cognitive workload, and redundancy versus complementarity. Across domains, haptic feedback was most consistently associated with immediate, localized, spatial, and directional information, whereas auditory feedback was particularly suitable for sequential, semantic, explanatory, and movement-related guidance. Visual feedback remained valuable for detailed, persistent, and spatially precise information when sufficient visual attention was available. Multimodal feedback showed the clearest benefits when channels conveyed complementary information, although additional modalities could also increase workload, task time, or synchronization demands. No single modality was universally optimal; effectiveness depended on task requirements, available sensory resources, environmental conditions, and feedback implementation. Based on these findings, the review proposes evidence-informed design guidelines and identifies longitudinal evaluation, real-world validation, personalization, context-aware adaptation, synchronization, and standardized reporting as priorities for future research.

Index Terms—multimodal feedback, eyes-busy interaction, hands-busy interaction, haptic feedback, auditory feedback, visual feedback, systematic literature review, human-computer interaction, context-aware systems, multimodal interaction

I. INTRODUCTION

Users regularly find themselves in situations where their visual and physical attention is divided between technology and primary duties as digital devices become increasingly embedded in everyday life. Driving, cooking, strolling, and engaging in physical activities are examples of “eyes- and hands-busy” situations that pose particular difficulties for human-computer interaction (HCI) design. Traditional interface models are insufficient when users are unable to rely on visual feedback or manual input, requiring alternative interaction modalities that can communicate information effectively without demanding additional cognitive or physical resources [1].

A promising approach to assisting users in these attention-constrained environments is multimodal feedback, which integrates two or more sensory channels, such as auditory, haptic, and visual cues [2]. In a variety of application domains, research has shown that adding tactile or auditory feedback to visual information can increase safety, improve task performance, and lessen cognitive workload [3, 4]. For instance, auditory cues can enhance spatial awareness and reaction times in demanding operational settings [5], while in-car interfaces that offer vibrotactile feedback have been demonstrated to decrease eyes-off-road time during secondary tasks [6].

Even with the increasing amount of evidence in favour of multimodal feedback, the current body of knowledge is still dispersed throughout various application domains, research communities, and journals. Accessibility researchers look into non-visual interfaces for people with visual impairments [7], wearable computing researchers study haptic feedback for fitness guidance [8], and automotive HCI researchers might study tactile alerts for collision avoidance [3]. It is challenging to synthesise findings and derive generalisable design

principles because these disparate fields use different terminologies, experimental methodologies, and evaluation metrics. As a result, designers and practitioners do not have a single set of evidence-based recommendations for using multimodal feedback in situations where hands and eyes are busy.

Research knowledge fragmentation poses a number of important challenges. First, designers might unintentionally rely on less-than-ideal feedback modalities or combinations in the absence of systematic synthesis, which could endanger user performance or safety. Second, it is difficult to determine which design elements consistently enhance results across contexts due to the abundance of studies that focus on specific feedback parameters, such as modality choice, timing, intensity, and location, without conducting a comparative analysis. Third, it is challenging for researchers to pinpoint knowledge gaps and set priorities for further research when there is a lack of consolidated evidence. These difficulties are especially severe in high-stakes applications where ineffective feedback could have detrimental effects.

A. Motivation and Objective

To meet these gaps, this systematic review summarizes research on multimodal feedback for hands-and-eyes-busy interaction over six major application domains: driving, cooking, walking with mobile devices, physical activity (including exercise and wearable devices), accessibility for users who are blind or have low vision (BLV), and surgery. These domains are comparable in that fundamental tasks occupy users' visual and/or manual attention, rendering traditional visual-only feedback or manual input modalities unsafe or ineffective. To identify frequently used multimodal feedback modalities, assess their efficacy in various interaction contexts, look at design elements that affect user performance and experience, gauge the strength of the available evidence, and identify unmet research gaps, the review methodically analyses empirical studies published across multiple application domains [9, 10]. In order to help researchers, designers, and developers create safer, more adaptive, accessible, and user-centred multimodal interactive systems for future hands-and-eyes-busy environments, this review presents evidence-informed design guidelines based on the synthesised findings [11, 8, 7].

B. Research questions

RQ1: What feedback modalities and combinations (auditory, haptic, visual, and multimodal) have been used in eyes- and hands-busy interaction, in which application domains, and with which interaction mechanisms (e.g., gaze, gesture, speech, or touch)?

RQ2: What evidence is reported for the effectiveness of different feedback modalities and combinations across eyes-busy and hands-busy contexts, and how do the observed patterns vary across application domains?

RQ3: What design factors (e.g., feedback timing, body location, redundancy vs. complementarity) consistently

improve user performance and experience across domains?

RQ4: How strong is the existing evidence behind these findings?

RQ5: What gaps remain in current research on multimodal feedback for eyes-/hands-busy interaction, and what should future studies investigate?

C. Positioning and Contributions

Existing reviews provide valuable but largely partial perspectives on multimodal interaction. Some focus broadly on multimodal human-computer interaction, while others concentrate on a particular sensory channel, technology, or user population, such as nonverbal auditory feedback, electrotactile interaction, haptic systems, or accessibility for blind and low-vision users [12, 11, 13, 14]. Consequently, evidence relevant to eyes- and hands-busy interaction remains distributed across application domains and is reported using different tasks, outcome measures, and feedback implementations.

The unique emphasis of the present review is thus not simply to catalogue multimodal technologies, but to examine how feedback should be designed when users have limited attentional resources. The review compares evidence across application contexts, identifies recurring design factors such as timing, feedback location, encoding, synchronization, and complementarity, and translates convergent findings into evidence-informed design recommendations. This cross-domain evidence-to-design synthesis distinguishes the present work from reviews organized primarily around individual modalities, technologies, or application areas.

The main contributions of this review are as follows:

- 1) We offer a methodical compilation of feedback for interactions that engage both visual and tactile senses across various application scenarios, categorising system-to-user feedback into auditory, haptic, visual, and multimodal types, while also examining interaction modalities like gaze, gesture, speech, and touch.
- 2) We analyse the circumstances in which various feedback modalities are useful, focusing specifically on information kind, task requirements, accessible sensory resources, and contextual application.
- 3) We recognise persistent design elements that affect the efficacy of feedback, such as timing, synchronisation, body placement, intensity, encoding, cognitive burden, redundancy, and complementarity.
- 4) We summarise the robustness, uniformity, and constraints of the existing evidence across several fields and pinpoint deficiencies related to long-term assessment, practical implementation, customisation, standardised metrics, accessibility, and adaptive feedback.
- 5) We translate the synthesized findings into evidence-informed design guidelines for developing context-

sensitive multimodal feedback systems in eyes- and hands-busy environments.

The remainder of this paper is organized as follows. Section III describes the review methodology. Section IV presents the modality-level results. Section V compares findings across application contexts. Section VI synthesizes recurring design factors, and Section VII translates the synthesis into evidence-informed recommendations. Section VIII discusses research gaps and future directions. Section IX presents the limitations of the review, and Section X concludes the paper.

II. BACKGROUND AND RELATED REVIEWS

Multimodal interaction has been studied from several perspectives, including sensory integration, multimodal interface design, assistive technologies, wearable interaction, and human-robot interaction. Broad reviews of multimodal HCI demonstrate that multiple sensory and interaction channels are increasingly used across domains such as healthcare, transportation, extended reality, and assistive computing [12]. However, such broad reviews are not specifically organized around situations in which users' visual and manual resources are constrained.

Other reviews provide deeper analyses of individual sensory channels. For example, research on nonverbal sound has examined the forms and communicative functions of auditory cues [11], while reviews of electrotactile and haptic technologies have examined tactile encoding, stimulation parameters, device characteristics, and applications [13, 15]. Accessibility research has additionally investigated how haptic technologies can support blind and low-vision users in navigation, graphical understanding, object interaction, and non-visual information access [14].

These reviews provide important modality-specific and domain-specific knowledge, but they do not fully resolve how feedback modalities should be selected and designed across different eyes- and hands-busy tasks. In particular, evidence about timing, spatial location, synchronization, information encoding, workload, and multimodal reciprocity remains distributed across separate research communities. The present review therefore focuses on synthesizing these design factors across contexts and translating recurring evidence into practical, evidence-informed feedback guidelines.

III. METHODOLOGY

In order to find, categorise, and compile data on system-to-user feedback in eye and hand-busy engagement, this study used a systematic literature review process. The review was organised in accordance with known guidelines for systematic reviews [9, 10]. Database searches, duplicate removal, title and abstract screening, full-text eligibility evaluation, structured data extraction, evidence coding, and cross-study synthesis were all part of the review process.

A. Information Sources and Search Strategy

Searches were conducted in IEEE Xplore, ACM Digital Library, Scopus, SpringerLink, and Elsevier ScienceDirect. Google Scholar was used as a supplementary source for citation chaining and keyword expansion. Search terms were organized into three conceptual groups: (1) attention-constrained interaction terms, (2) feedback modality terms, and (3) interactive-system terms. Terms within each group were connected using OR, and the groups were combined using AND.

The search vocabulary included attention-related terms such as “eyes-busy”, “hands-busy”, “eyes-free”, and “hands-free”; feedback terms such as “multimodal feedback”, “auditory feedback”, “audio feedback”, “haptic feedback”, “tactile feedback”, “vibrotactile feedback”, “visual feedback”, and “cross-modal feedback”; and interaction terms such as “human-computer interaction”, “user interface”, “wearable interaction”, and “interactive system”.

A representative search structure was:

(“eyes-busy” OR “hands-busy” OR “eyes-free” OR “hands free”) AND (“multimodal feedback” OR “auditory feedback” OR “audio feedback” OR “haptic feedback” OR “tactile feedback” OR “vibrotactile feedback” OR “visual feedback” OR “cross-modal feedback”) AND (“human-computer interaction” OR “user interface” OR “wearable interaction” OR “interactive system”)

We adapted database-specific syntax to each source's search interface while preserving the same conceptual search structure.

B. Eligibility Criteria

Studies were considered eligible when they examined system-to-user feedback in an interaction context where visual attention, manual resources, or both were constrained by a core task. Relevant feedback included auditory, haptic, tactile, vibrotactile, visual, and combinations of these channels. Studies were required to report findings relevant to interaction performance, accuracy, workload, usability, user experience, safety, feedback recognition, or feedback design.

Studies that only focused on sensing, recognition, or user input were not considered feedback evidence unless they also assessed the data that the system provided to the user. In order to distinguish feedback modalities from interaction mechanisms like gaze, gesture, speech, or touch, this distinction was used.

C. Study Selection

Duplicate records were eliminated and records gathered from the chosen databases were integrated. After that, titles and abstracts were compared to the predetermined eligibility requirements.

Each study was assessed for (1) relevance to an eye- or hands-busy interaction context, (2) the presence of

TABLE I: Eligibility criteria for study selection

Category	Inclusion Criteria	Exclusion Criteria
Interaction Context	Eyes-busy, hands-busy, eyes-free, hands-free, or closely related attention-constrained interaction.	No relevant visual, manual, or attentional constraint.
Feedback	System-to-user auditory, haptic, tactile, vibrotactile, visual, or multimodal feedback.	Sensing, classification, or user-input systems without evaluated system-to-user feedback.
Application Scope	Relevant interactive contexts including transportation, domestic tasks, physical activity, accessibility, mobile interaction, XR, healthcare, teleoperation, or other attention-demanding tasks.	Tasks unrelated to the review questions or without a relevant constrained-interaction context.
Outcomes	Performance, accuracy, error, response time, workload, usability, safety, user experience, recognition, preference, or feedback-design findings.	No outcome or design evidence relevant to feedback.
Language	English-language publications.	Publications not available in English.
Publication Period	Studies published from 2005 to the final search date in 2026.	Studies published before 2005, except when cited only as background or theoretical context.

system-to-user feedback, (3) an eligible study and publication type, and (4) the availability of outcomes relevant to the research questions after potentially relevant records moved on to full-text assessment.

Reasons for exclusion were noted during full-text screening in order to preserve an auditable selection procedure. Studies that solely focused on sensing, recognition, or user input were not included in the feedback evidence base unless they also assessed the data that the system provided to the user.

The search identified 161 records. After removal of 28 duplicate records, 133 records remained. A further 23 records were excluded because they fell outside the predefined publication period, leaving 110 records for title and abstract screening. During title and abstract screening, 19 records were excluded as not relevant to the review criteria, and 91 reports proceeded to full-text eligibility assessment. Following full-text assessment, 22 reports were excluded because they did not meet the eligibility criteria. The final synthesis included 69 unique publications. Figure 1 summarizes the complete selection process.

D. Data Extraction and Coding

Each qualified study's data was recorded using a standardized extraction form. Publication year, application domain, interaction context, participant population, study type, sample size, evaluation method and measures, feedback modality, interaction mechanism, principal findings, reported quantitative outcomes, design implications, evidence limitations, and study-specific challenges were among the variables that were extracted.

Feedback was coded according to the system-to-user sensory channel. Auditory feedback included speech, sonifi-

cation, and non-speech sounds. Haptic feedback included vibrotactile, electrotactile, pressure, force or kinesthetic, skin-stretch, and related tactile forms. Visual feedback included graphical displays, overlays, augmented-reality cues, visual system states, and video-based guidance. Publications combining two or more system-to-user sensory channels were classified as multimodal. Feedback modality was coded separately from user-input mechanisms such as gaze, gesture, speech input, and touch input.

E. Evidence Assessment

Evidence strength was assessed at the level of synthesized findings rather than being determined solely by statistical significance in an individual study. The assessment considered the number of relevant supporting studies, consistency of the reported pattern, directness of the evidence to the review question, study-design limitations, sample characteristics, and the presence of contradictory findings.

For synthesis-level interpretation, evidence was classified as *Strong*, *Moderate*, or *Limited*. Strong evidence required at least three directly relevant studies reporting generally consistent findings. Moderate evidence was assigned when a pattern was supported by two relevant studies or when multiple studies reported mixed or heterogeneous results. Limited evidence was assigned when a conclusion relied primarily on a single study, exploratory evidence, an indirect application context, or evidence transferred from an adjacent domain. Secondary reviews were used to contextualize and corroborate patterns but were not counted as independent empirical replications of the primary studies they summarized.

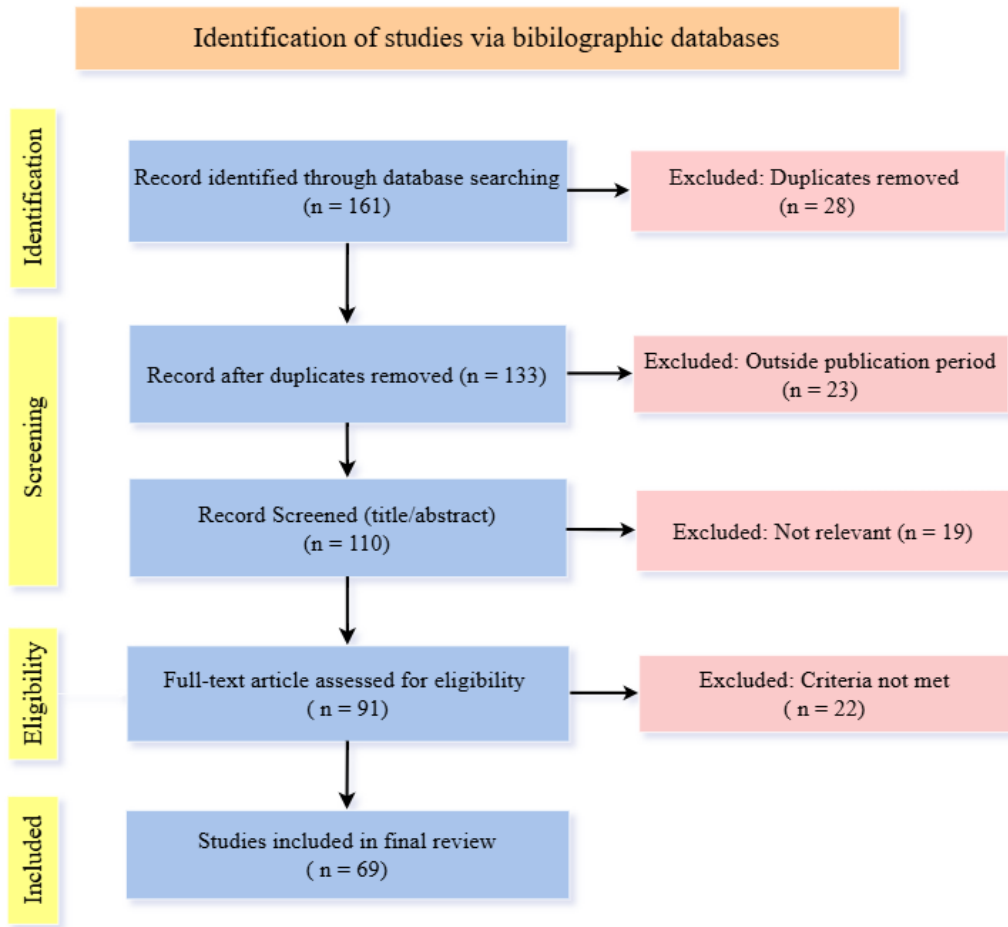


Fig. 1: PRISMA diagram of study selection.

F. Synthesis Procedure

A pooled meta-analysis was deemed inappropriate due to the significant differences in participant populations, tasks, feedback implementations, experimental designs, and outcome measures among the analyzed research. As a result, a structured narrative synthesis was employed. Three phases of the synthesis were carried out. To find recurrent strategies, advantages, and drawbacks, research were first categorized by feedback mode. Second, results were examined across application contexts to see if task requirements and user resource availability affected efficacy. Third, recurrent design elements were identified from several investigations, such as time, feedback location, intensity and encoding, synchronization, cognitive effort, redundancy, complementarity, and user-specific requirements.

When they helped with interpretation, quantitative results from individual studies were kept; however, numerical values derived from various tasks and outcome measures were not regarded as directly comparable cross-study effect estimates.

IV. RESULTS

This section presents the findings of the review across auditory, haptic, visual, and multimodal feedback. The analysis first summarizes the characteristics of the evidence base and then examines modality-specific techniques, reported outcomes, application contexts, design implications, and recurring limitations. For presentation, each included publication was assigned to one primary feedback category according to the system-to-user feedback configuration emphasized in the review. Publications combining two or more system-to-user sensory channels were categorized as multimodal.

The following subsections examine auditory, haptic, visual, and multimodal feedback, focusing on the techniques evaluated, reported outcomes, application contexts, design implications, and limitations.

A. Overview of the Evidence Base

Following the final eligibility assessment, 69 unique publications were retained for synthesis. The literature covered multiple application contexts, including transportation, domestic and procedural tasks, physical activity

and wearable interaction, accessibility, mobile interaction, extended reality, and precision or high-stakes tasks. Study designs ranged from controlled experiments and simulator evaluations to in-home studies, observational research, and other user evaluations.

The evidence base included auditory, haptic, visual, and multimodal feedback approaches. Because individual studies could examine more than one feedback channel, modality categories were treated as non-exclusive where appropriate.

B. Evidence Assessment of Synthesized Findings

The number of directly relevant studies, the consistency of findings across studies and domains, and the methodological and reporting constraints of the available information were all taken into account when evaluating the strength of the evidence at the synthesis level. Evidence was categorized as Strong when multiple directly relevant studies showed broadly consistent findings, Moderate when evidence was supported by fewer studies or showed some inconsistency, and Limited when findings were primarily supported by a single study or indirect evidence. These categories should not be taken as general rankings of feedback modalities; rather, they indicate the strength of the evidence provided for each synthesized pattern. Table II summarizes the resulting assessment of the major cross-study findings.

C. Audio Feedback

Users can get advice without constantly depending on visual attention thanks to audio feedback, a crucial interaction mode that conveys information through spoken instructions, conversational dialog, nonverbal sounds, and sonification. Instructional support, cooking, human-robot interaction, physical activity, accessibility, and context-aware interaction have all benefited from its use [40, 41, 19, 11, 20, 8]. Through speech, sound patterns, and pitch variations, various types of audio feedback can provide task guidance, communicate system statuses, give contextual information, and indicate movement or activity changes [19, 11, 8]. In circumstances where continuous visual engagement may be uncomfortable or impracticable due to eye or hand fatigue, audio feedback is very helpful [40, 20]. However, its effectiveness can be affected by speech-recognition errors, response latency, background noise, excessive verbal information, conversational interruptions, and cognitive workload [40, 19, 42]. Therefore, effective audio feedback requires careful consideration of information structure, timing, interaction context, and the user's cognitive demands.

Huh et al. [40] suggested a voice-activated, interactive assistant to improve online educational videos' visual accessibility. Using Whisper for automatic speech recognition (ASR), GPT-4 for taxonomic categorization, and automated completion criteria generation, they divided

video transcripts into organized, step-by-step action objects in order to turn videos into interactive voice assistants. They attempted to address the drawbacks of using human-powered visual interpreting services (like AIRA and BeMyEyes) during manual tasks, including the high fees, connection delays, lack of work knowledge, and hand-busy limitations. Eight blind and low-vision (BLV) participants watched 14 instructional videos as part of their user evaluation. They discovered that although the interactive format was very helpful, there was no statistically significant change in the participants' ratings of the feedback itself. ($\mu = 1.28$, $\sigma = 4.75$ for Vid2Coach vs. $\mu = 1.28$, $\sigma = 5.25$ for the baseline; $Z = 0.85$; $p > 0.05$). Key limitations of this approach include the camera often missing user mistakes due to a narrow field-of-view or occlusions, speech-recognition failures under background kitchen noise, and intent classification errors (such as misinterpreting "It's fun" as "It's done" and prematurely advancing to the next step).

Chaves and Gerosa [41] examine how conversational voice assistants integrate into natural, multi-party household conversations. Their approach utilized a Wizard of Oz (WOZ) experimental simulation where human operators ("wizards") selected pre-scripted chatbot responses based on keyword searches of a spreadsheet database, with the constraint that they could not adapt their vocabulary or composition. In order to comprehend users' natural expectations of conversational turn-taking and sequential coherence without technological system failures, they attempted to resolve and get beyond natural language processing (NLP) restrictions. By transforming the conversational corpora into 24 numerically similar observations, their findings effectively mapped these dynamics and revealed no statistically significant variations in turn-taking dynamics between single- and multi-chatbot settings. The artificial typing latency of 5–10 seconds introduced by human wizards, which cannot replicate computational speeds, potential subjectivity bias during qualitative analysis, and a small, unrepresentative sample of students with STEM backgrounds around the same age are the main limitations of this approach.

Hwang et al. [19] established design guidelines to transform written instructions into audio-friendly formats. Their approach involved an in-home observational study of 12 participants cooking recipes of their choice while being step-guided by an audio-only Amazon Alexa speaker. The study addressed cognitive and instructional-tracking difficulties that arise when written instructions are delivered verbatim through voice interfaces. Long or poorly structured spoken instructions can increase working-memory demands and make it difficult for users to retain their place within a procedure. Their results identified 9 core user-experience challenges and proposed a framework of 8 distinct HCI rewriting goals (such as Summarize, Signpost, Split, and Reorder), achieving high inter-rater coding reliability across conversational

TABLE II: Evidence Assessment of Major Synthesized Findings

Synthesized Finding	Relevant Contexts	Evidence Pattern	Evidence Strength	Main Limitations
Haptic feedback for localized, spatial, and directional information	Driving, physical activity, accessibility, mobile interaction, precision tasks	Multiple studies associate haptic feedback with immediate, localized, spatial, or directional guidance. Examples include reduced interaction time, improved non-visual navigation, and improved spatial or alignment performance [6, 16, 17, 18].	Strong	Different haptic technologies, body locations, intensities, and task conditions limit direct comparison.
Auditory feedback for semantic and sequential information	Cooking, physical activity, rehabilitation, operational and mobile interaction	Auditory and speech-based feedback is repeatedly used for sequential instructions, explanations, procedural guidance, and continuous movement information [19, 20, 21, 22].	Strong	Speech content, timing, environmental noise, cognitive workload, and interaction delays vary substantially across studies.
Visual feedback when visual attention is available	Rehabilitation, XR, industrial guidance, surgery and precision interaction	Visual and AR feedback supports detailed, persistent, and spatially precise information when users can allocate sufficient visual attention to the interface [23, 24, 25, 26].	Moderate	Visual feedback can increase workload, occlusion, completion time, and attentional demand when visual resources are constrained [27, 28, 29].
Complementary multimodal feedback	Driving, navigation, XR, mobile interaction, teleoperation, precision tasks	Combining channels can improve performance or usability when different modalities communicate complementary information rather than simply duplicating the same signal [30, 31, 32, 18, 33].	Strong	Multimodal systems can introduce additional workload, learning requirements, task time, and synchronization demands [34, 35, 36].
Feedback timing and temporal synchronization	Driving, cooking, mobile interaction, teleoperation, physical interaction	Studies indicate that delayed feedback can disrupt interaction and that temporal mismatch between channels can reduce the quality of multimodal feedback. Immediate feedback can improve interaction efficiency in some tasks [20, 16, 36, 37, 30].	Strong	Reported timing requirements are task- and modality-dependent; evidence does not establish a universal synchronization threshold.
Feedback location and spatial correspondence	Haptic guidance, accessibility, physical activity, precision interaction	Localized feedback is repeatedly associated with information that refers to a specific body region, direction, or spatial relationship [17, 18, 38, 13].	Moderate	Body location, device type, task movement, and user characteristics differ across studies.
Information density and cognitive workload	Driving, XR, multimodal interaction, rehabilitation	Increasing the amount or complexity of feedback can increase workload or task time, whereas appropriately distributed or progressive feedback may reduce perceived effort [39, 27, 35, 34].	Moderate	Workload measures and experimental conditions are heterogeneous, limiting direct cross-study comparison.
Redundant multimodal feedback is not universally beneficial	Driving, XR, mobile and high-stakes interaction	Additional channels may provide benefits, but redundant information can also increase workload, synchronization demands, or task duration. Benefits are more consistently reported when channels have distinct complementary roles [34, 35, 39, 31].	Moderate	Evidence is heterogeneous and depends strongly on task complexity, modality implementation, and user resources.
Context-dependent modality selection	Across all reviewed domains	The usefulness of a modality varies according to information type, task demands, available sensory resources, environmental conditions, timing, and implementation. No universal modality hierarchy is supported [6, 16, 19, 26, 18].	Strong	Cross-domain studies use different populations, tasks, measures, and experimental designs, preventing direct quantitative comparison.

turns with a Krippendorff's α ranging from 0.80 to 1.00. The primary limitations of their work include Alexa's frequent failure to listen or respond, which forced an average user wait time of 4.6 seconds with a wake word, and a narrow demographic of young, college-educated adults that does not represent safety-critical high-stakes environments.

Zhang and Fitter [11] conducted a systematic review of nonverbal sound in human-robot interaction. The review screened 3,799 records and included 148 publications from 1996 to April 2022. The study addressed terminological fragmentation and the lack of a common framework for describing nonverbal auditory cues in human-robot interaction.. Their approach involved a scrupulous

systematic review of peer-reviewed articles, screening 3,799 search results to analyze a final set of 148 relevant studies published between 1996 and April 2022. The system was designed to address the severe terminology fragmentation, jargon overlap, and lack of shared design taxonomies across the fields of social robotics and auditory displays. The review developed a taxonomy describing nonverbal sounds according to their form and communicative function. However, the limitations of their approach include finding that 20.2% of the analyzed systems in the original dataset were purely experimental, the potential of novelty effects in the 60% of studies conducted in-person in laboratory settings, and the loss of original audio samples due to link rot over time.

Jaber et al. [20] investigated how proactive, context-aware voice assistance influences human-machine collaboration during complex cooking tasks. Their approach used a Wizard of Oz (WOZ) setup where a human wizard watched 16 participants cook via a camera feed and set off Google text-to-speech responses on a Mycroft.ai smart speaker. The study addressed interaction breakdowns caused by assistants that lack sufficient awareness of the user's current task state and context. The analysis showed that context-aware, proactive voice assistants significantly improve task success and reduce dialogue repair loops, though the system's response time introduces a 2–5-second delay in multimodal fusion. The main limitations of their approach include the inherent bandwidth limits of spoken language in human-computer interaction, conversational friction or user frustration if proactive interruptions are timed poorly, and privacy concerns associated with collecting continuous task context.

Chan et al. [42] proposed “Mango Mango”, an LLM-based conversational assistant designed to evaluate whether Large Language Models can function as adaptive task partners. The system used GPT-3.5-Turbo to power an Alexa skill (which included recipe suggestions and conversation history logs) and assessed it using a 12-person in-home study. They attempted to address the limitations of conventional, heuristic-based voice assistants, such as their strict conversational templates, lack of contextual reasoning, and incapacity to handle free-form queries. With an overall trustworthiness score of 4.11/5 on the Voice Usability Scale (VUS) and low NASA-TLX task workload ratings, their results showed good usability and satisfaction scores. However, GPT-3.5-Turbo's vulnerability to semantic errors (such as misinterpreting the “last instruction” as the last step of the entire recipe rather than the “previous” step) and back-and-forth dialog increasing cognitive load, which presents safety risks near active hot stoves, are the limitations of their approach.

Singh et al. [8] introduced a body-effect movement sonification framework intended to influence body perception changes and encourage physical activity. The system consisted of a custom smartphone application worn on the participant's thigh, which monitored joint

angles and generated continuous 1.3-second consonant pitch shifts (ascending and descending musical tones) in real time, evaluated across a remote study ($n = 22$) and an in-person laboratory study ($n = 21$) with no-feedback and haptic control conditions. They aimed to solve psychological and physical barriers to strength training—such as fatigue, perceived difficulty, and lack of motivation. Their results showed that ascending pitch significantly increased feelings of lightness ($F(1, 21) = 13.21, p = 0.002, \eta_p^2 = 0.39$), motivation ($F(1, 21) = 13.63, p = 0.001, \eta_p^2 = 0.39$), capability ($F(1, 21) = 5.30, p = 0.02, \eta_p^2 = 0.23$), and decreased perceived tiredness ($F(1, 21) = 7.64, p = 0.012, \eta_p^2 = 0.27$); conversely, descending pitch triggered feelings of heaviness but significantly increased actual physical maximum acceleration ($F(1, 21) = 6.20, p = 0.021, \eta_p^2 = 0.23$). The author noted several limitations of their work, including the use of female-only participants in both experiments to reduce physiological confounding, “sound blurring” sensory overload during rapid squats where pitch-shift durations exceeded movement times, and minor device-specific audio latencies in some smartphone models that disrupted participants' sense of agency.

Kantan et al. [43] developed a real-time, inertial sensor-based concurrent auditory biofeedback system using four inexpensive wireless IMUs (M5Stack Gray) attached to the thighs and shanks to sonify lower-limb segment angular velocities during the swing phase. Their goal was to address hemiparetic gait impairments—specifically swing-phase kinematic asymmetries and low walking cadence—in stroke and brain injury patients. Their approach converted leg angular velocities into ecological “wading” water sounds, enhanced with splash sounds for positive reinforcement upon reaching target velocity thresholds and urinating sounds for negative reinforcement at low swing speeds. In a feasibility evaluation with 9 hemiparetic patients and 7 physiotherapists, 3 patients exhibited immediate kinematic improvements, including increased cadence, better knee deceleration control, and enhanced segment angular velocity symmetry. However, the evaluation was limited by its small sample and the absence of a randomized controlled trial. The system also had difficulty identifying subtle gait asymmetries in faster walkers, while the combination of multiple auditory layers could create cognitive clutter. Furthermore, the required software and hardware configuration was considered too complex for independent clinical use by physiotherapists.

Douglass-Kirk et al. [21] created the Sonic Sleeve, an automated digital biofeedback system using a 2D webcam, OpenPose computer vision, and Wekinator machine learning to track upper-limb kinematics (neck, wrist, elbow, shoulder) during active reaching tasks. They aimed to reduce compensatory abnormal movements (such as trunk flexion, shoulder abduction, and shoulder elevation) in chronic stroke patients without using restrictive

physical harnesses. Their approach provided real-time binary auditory feedback by muting self-selected music whenever abnormal movements exceeded therapist-calibrated thresholds. In a study of 20 chronic stroke patients, the duration of abnormal movement was significantly reduced from 39.4% without feedback to 19.3% with feedback ($F(1, 18) = 9.424, p = 0.007$, partial $\eta^2 = 0.344$), representing a 20.1% absolute reduction, with 14 out of 20 patients showing statistically significant individual improvements. The system nevertheless had several limitations. Its single binary mute signal did not indicate which compensatory movement had triggered the feedback, and 3 participants experienced an increase in abnormal movement duration. In addition, the system was not evaluated for long-term use in home settings, and baseline clinical characteristics that might explain differences between responders and non-responders were not examined.

de Jesus Oliveira et al. [44] developed a wearable medical-grade smart insole system equipped with textile pressure sensors and 3DOF accelerometers to provide automated gait disorder classification and auditory biofeedback. They sought to assist everyday users during static and dynamic physical therapy exercises and tel-rehabilitation by detecting functional gait disorders and knee malalignments (varus/valgus). Their approach implemented a firmware-embedded Feed-Forward Neural Network (FFNN) to estimate Ground Reaction Forces (GRFs) and a Support Vector Machine (SVM) trained on the GaitRec dataset to classify physiological vs. pathological gait, triggering event-based guitar chimes or continuous wind/trumpet sounds upon state deviations. Their system achieved a low normalized root mean square error (NRMSE) for vertical GRF (3.95%) and anteroposterior GRF (3.81%), a 91% SVM classification accuracy on test data (97% on estimated GRFs), and a “Good” System Usability Scale (SUS) score of 74 among 10 users. Limitations included lower accuracy in estimating mediolateral GRF (9.49% NRMSE). Validation was also conducted mainly with healthy participants and expert focus groups rather than individuals with gait disorders. In addition, some users found the trumpet feedback intrusive compared with the wind sound.

Linnhoff et al. [45] presented *sofigait*, a wireless prototype measuring real-time sagittal knee angle kinematics using four MIMUs (Bosch BNO080) on the thighs and shanks paired with a tablet sonification module. They aimed to provide a portable, cost-effective tool to help patients perceive spatiotemporal knee joint asymmetries (e.g., limping) during gait rehabilitation. Their approach used parameter-mapping sonification to translate knee joint trajectories into continuous sine tones varying in pitch and volume over four experimental sound conditions (testing swing vs. stance accentuation and high vs. low pitch). Validated against a 3D Vicon optical capturing system with 22 participants, the system yielded

root-mean-square errors (RMSE) of $7.6^\circ \pm 2.6^\circ$ (left) and $6.9^\circ \pm 3.1^\circ$ (right), while a sound perception trial with 28 participants showed that lower pitch significantly improved pleasantness ($F = 9.22, p = 0.005$), making the low-pitched swing-accentuated version (Soni 1.2) the most preferred (information score: 4.7 ± 1.2 , pleasantness score: 4.3 ± 1.4). Limitations included systematic overestimation of peak knee flexion (mean bias up to $-7.52^\circ \pm 6.2^\circ$, with statistical parametric mapping showing significant differences over 40–50% of the gait cycle). Furthermore, the evaluation used healthy participants with artificially restricted knee movement rather than clinical patients, and measurement accuracy could be affected by sensor alignment errors and drift.

Weller et al. [22] proposed Auditory Step Feedback Redirected Walking (ASRDW), a virtual reality (VR) redirection system incorporating pressure-sensor shoe soles connected to an Arduino for footstep detection. They aimed to redirect users subconsciously while walking toward a VR target so they stay within physical room boundaries without noticing path manipulation. Their approach played spatialized footstep sounds mapped to user position (gravel for straight paths, changing to water or wet lawn warnings to induce corrective turning) and combined this with eye-blink visual redirection (VRDW). In a study with 20 participants in a 10×20 m area, pure ASRDW achieved a statistically significant positive translational redirection of 1.7 m (median 205 cm vs. 10 cm baseline, $p < 0.001$), while combining ASRDW with eye-blink VRDW increased redirection to 2.5 m (median 235 cm, $p < 0.001$), with 16/20 and 18/20 participants remaining completely unaware of the redirection, respectively. Limitations included testing only translational deviation in a dark, visually minimal VR environment, no evaluation of rotational gains or smaller room scales, observed gender disparities (men were significantly more influenced than women), and reliance on unpleasant warning sounds (water/mud) that may not fit all VR narrative contexts.

Peyre et al. [46] developed an upper-limb motion sonification setup using wireless IMU motion sensors on the arm to evaluate arm sliding extension-flexion movements. They aimed to systematically investigate both the quantitative temporal effects and qualitative user experiences of different sound feedback categories in post-stroke hemiparetic patients versus healthy controls. Their approach evaluated 5 sound types across 3 categories (direct sound modulation: Pitch, Drum; melody playing: Discrete, Continuous; environmental soundscape: nature sounds) against a no-sound control condition during unconstrained-speed arm sliding. Across 15 hemiparetic patients and 15 matched healthy controls, normalized total movement duration was significantly longer with sonification compared to no-sound ($p < 0.001$ for both groups and arms), driven by prolonged holds in maximum extension; qualitatively, 29 of 30 participants preferred

moving with sound, rating the Soundscape highest (59/75 points for patients, 57/75 for healthy) and describing the experience as “playful” (30/75 for patients, 48/75 for healthy). Limitations included a single-session evaluation without longitudinal motor relearning assessments, a small sample size ($N = 15$ per group) with high heterogeneity in motor impairment among stroke patients, and a non-randomized presentation order of sound conditions.

D. Haptic Feedback

By communicating information through tactile sensations including vibration, pressure, force, and skin stimulation, haptic feedback is a vital interaction mode that enables users to receive information without mainly depending on visual or aural attention. Directional navigation, spatial awareness, gesture interaction, virtual reality, accessibility, rehabilitation, and emotional communication have all benefited from its application [47, 13, 14, 15]. Design elements that affect users’ perception and interpretation of tactile cues, such as intensity, duration, frequency, waveform, rhythm, spatial placement, and temporal pattern, determine how effective haptic feedback is. [47, 16, 48]. Additionally, haptic feedback can lessen reliance on constant visual attention and offer supplementary information during visually demanding or cognitively taxing tasks [49, 50]. However, its usefulness and efficacy may be impacted by issues with hardware complexity, calibration, spatial and temporal resolution, recognition accuracy, cognitive workload, user variability, and system responsiveness [13, 51]. Therefore, careful consideration of tactile characteristics, stimulation location, timing, task context, and user needs is necessary to design efficient haptic feedback

Haghighi et al. [47] proposed a systematic evaluation of how five core haptic properties (amplitude, waveform, duration, rhythm, and spatio-temporal pattern) behave ambiently. Their methodology included a forearm-mounted vibrotactile band in conjunction with a primary delayed digit recall task (the MIT AgeLab 1-back test) to simulate the auditory and cognitive demands of everyday multi-tasking. They attempted to tackle the absence of empirical standards for ambient haptic displays, given that past haptic investigations have mainly concentrated on isolated settings where a user’s attention is entirely dedicated to the haptic device. Their quantitative results across 16 participants showed that amplitude and duration are the most effective channels under cognitive load, yielding the highest gradient-detection accuracy with median scores of 38.000 and 39.000 correct trials out of 40 (Amplitude mean: 37.063 ± 4.582 ; Duration mean: 37.125 ± 3.612), with amplitude imposing the lowest cognitive workload. The primary limitations of their approach are that rhythm took significantly longer to identify than duration, the spatial-temporal pattern was complex to decode, and subjective NASA-TLX workload ratings (measured out of 7) revealed that female and non-native populations

reported significantly higher subjective stress and mental load.

Kourtesis et al. [13] proposed a technological overview and systematic review mapping the state-of-the-art in electrotactile feedback applications. Their approach applied the PRISMA framework to filter 323 candidate studies down to 78 peer-reviewed articles from 2010 to 2021 to analyze tactile and kinesthetic parameter modulations (current, pulse width, frequency, and spatial configurations) on the upper limbs. The review examined technological limitations affecting electrotactile feedback adoption, including wearability, device size, hardware cost, power consumption, and the need for user-specific stimulation calibration. The review reported that thin, flexible epidermal electrode systems such as Tacttoo achieved spatial resolutions as low as 1.15 mm, compared with approximately 4–6 mm for conventional mechanical pin displays. Key limitations of this approach are that electrotactile stimulation requires recurrent, personalized calibration to remain comfortable and safe as skin bio-impedance changes, and 66% of the literature consists of small pilot studies $N \leq 10$, while only 18% achieved a statistically robust sample of $N \geq 15$.

Jiang et al. [14] proposed a comprehensive review of haptic assistive tools, categorizing device structures, feedback modalities, and on-body stimulation locations. Their approach synthesized twenty years (2004–2024) of HCI accessibility literature using thematic clustering and inductive coding across 11 premium journals and conferences. The review addressed fragmentation in assistive haptic-system design by developing a unified framework linking haptic mechanisms, body locations, and application tasks. The review found that pressure was the most frequently reported feedback type (79/132 publications), followed by kinesthetic feedback (36/132) and vibration (32/132). Pressure was frequently used for graphical and map understanding, whereas vibration and kinesthetic feedback were commonly applied to navigation and warning tasks. Key limitations include the fact that kinesthetic devices offer high spatial precision (11% Weber Fraction error sensitivity) but are heavy, portable skin-stretch options are lightweight but suffer from low resolution (35% Weber Fraction sensitivity), and severe hardware constraints (the noise and conspicuous appearance of devices trigger user stress and social stigma).

Bouzbib et al. [15] proposed a dual-dimension design space to classify VR haptic systems based on their degree of physicality (from passive props to virtual objects) and degree of actuation (from no actuation to autonomous robotic displacements). Their approach analyzed the interactive workflows of current haptic devices—specifically comparing CoVR, ShapeShift, Snake Charmer, and Beyond the Force (BTF) drones—to balance user experience and hardware complexity. The study addressed difficulties in comparing heterogeneous VR haptic systems by organizing them within a com-

TABLE III: Summary of Audio Feedback Studies

Ref.	Audio Technique	Key Finding / Quantitative Evidence	Contexts / Applications	Limitations
Huh et al. [40]	Interactive voice guidance using ASR and LLM-generated step-by-step instructions.	Evaluated with 8 BLV participants across 14 videos. Feedback ratings did not differ significantly from baseline ($Z = 0.85$, $p > 0.05$), although real-time guidance supported task completion.	Accessible instructional videos and hands-busy procedural guidance for BLV users.	Small sample; camera occlusion, noise, ASR errors, and intent-classification failures.
Chaves and Gerosa [41]	Wizard-of-Oz conversational voice interaction using scripted agent responses.	Converted conversations into 24 comparable observations; no significant differences in turn-taking dynamics between single- and multi-agent conditions.	Multi-party household conversation, smart-home assistants, and conversational agents.	Artificial 5–10 s response delay; small, homogeneous participant sample.
Hwang et al. [19]	Step-by-step audio instructions delivered through Amazon Alexa.	Identified 9 recurring interaction problems and 8 rewriting goals; coding reliability was high ($\alpha = 0.80$ – 1.00).	Hands-busy cooking and procedural voice guidance.	Long instructions can overload working memory; Alexa failures caused interaction delays.
Zhang and Fitter [11]	Nonverbal sounds including electronic cues, vocables, localization, and expressive audio.	Screened 3,799 records and synthesized 148 studies, producing a taxonomy linking sound form to localization, recognition, and expression.	Human–robot interaction and nonverbal robot communication.	Heterogeneous, largely laboratory evidence; novelty and environmental noise limit generalization.
Jaber et al. [20]	Context-aware proactive voice assistance using task-state information.	Reduced communication breakdowns and dialogue-repair loops during cooking; multimodal fusion introduced approximately 2–5 s delay.	Cooking and high-workload domestic multitasking.	Processing delay and poorly timed interruptions can disrupt interaction; privacy concerns remain.
Chan et al. [42]	LLM-based conversational assistant using contextual prompts and dialogue history.	Reported high task success ($4.50 \pm 0.52/5$), satisfaction ($4.33 \pm 0.89/5$), trust ($4.25 \pm 0.87/5$), and efficiency ($4.00 \pm 1.35/5$).	In-home cooking and extended conversational task assistance.	Small sample ($n = 12$); semantic errors and repeated dialogue can increase cognitive load.
Singh et al. [8]	Continuous movement sonification using ascending and descending pitch.	Ascending pitch increased lightness ($F = 13.21$, $p = .002$), motivation ($F = 13.63$, $p = .001$), capability ($F = 5.30$, $p = .02$), and reduced tiredness ($F = 7.64$, $p = .012$); descending pitch increased maximum acceleration ($F = 6.20$, $p = .021$).	Exercise motivation, body perception, and strength training.	Female-only cohorts; rapid movement caused sound blurring; device latency sometimes disrupted agency.
Kantan et al. [43]	Ecological sonification mapping leg angular velocity to continuous water sounds.	3/9 patients showed immediate gait improvements; 6/9 rated the feedback fun and 7/9 natural-sounding.	Hemiparetic gait training after stroke or brain injury.	Small non-RCT sample; auditory clutter; some cues felt artificial; subtle asymmetries were difficult to detect.
Douglass-Kirk et al. [21]	Binary auditory biofeedback that mutes music when compensatory movement exceeds thresholds.	Abnormal movement decreased from 39.4% to 19.3% ($F(1, 18) = 9.424$, $p = .007$, partial $\eta^2 = .344$), a 20.1% absolute reduction; 14/20 improved significantly.	Chronic-stroke upper-limb rehabilitation.	One mute cue could not identify the specific error; 3/20 worsened; no long-term home evaluation.
de Jesus Oliveira et al. [44]	Hybrid sonification using guitar chimes and continuous wind/trumpet sounds.	GRF NRMSE was 3.95% vertical and 3.81% anteroposterior; SVM accuracy was 91% on test data and 97% on estimated GRFs; SUS=74/100.	Tele-rehabilitation, physical therapy, and gait/knee-alignment training.	Mediolateral error was higher (9.49%); trumpet sounds were poorly rated; mainly healthy participants were tested.
Linnhoff et al. [45]	Continuous knee-angle sonification using pitch- and volume-varying sine tones.	RMSE was $7.6^\circ \pm 2.6^\circ$ left and $6.9^\circ \pm 3.1^\circ$ right. Lower pitch improved pleasantness ($F = 9.22$, $p = .005$); Soni 1.2 received the highest ratings.	Gait training with artificially restricted knee flexion.	Peak knee flexion was overestimated; healthy participants with braces rather than clinical patients were tested.
Weller et al. [22]	Spatialized footstep sounds triggered by shoe sensors for redirected walking.	Audio-only produced about 1.7 m redirection (205 cm vs. 10 cm, $p < .001$); audio+visual reached about 2.5 m (235 cm, $p < .001$). 16/20 and 18/20 remained unaware.	Redirected walking in VR.	Minimal VR environment; no rotational gains; gender differences; warning sounds may not suit all narratives.
Peyre et al. [46]	Five upper-limb sonification designs using pitch, drums, melodies, and soundscapes.	Movement duration increased with sonification ($p < .001$); 29/30 preferred sound. Soundscape ranked highest (59/75 patients; 57/75 healthy).	Upper-limb movement in hemiparetic patients and healthy controls.	Single session; small groups ($N = 15$ each); heterogeneous impairment; fixed non-randomized sound order.

mon design space. The resulting taxonomy identified encountered-type haptic devices as a flexible compromise between physical realism, actuation capability, and interaction complexity. Key limitations of this approach include the prohibitive hardware cost of force-feedback devices (ranging from \$150 to \$20,000), the wind, noise, and physical safety hazards of actuated propellers, and the severe lack of standardized haptic evaluation protocols across the virtual reality literature.

Xu et al. [16] proposed a gesture-haptic UI for wearable shortcut commands, defining optimal vibrotactile pattern

designs for sequential hand gestures. Their approach compared “immediate feedback” (triggering haptic patterns instantly after each gesture) with “concatenated feedback” (queueing all patterns at the end of the sequence) using a wristband with 4 linear resonant actuators (LRAs) vibrating at 170 Hz. The system was designed to address sluggish, disjointed communication loops in user-initiated sequential inputs, defining a threshold where pattern durations can be shortened without sacrificing accuracy. Their results showed that durations of 0.3 s to 0.5 s achieve high recognition accuracy (81% to

89% medians), and the immediate feedback strategy is significantly faster and more efficient (reducing overall interaction time from 11.58 s to 8.56 s for 0.3 s duration) while lowering cognitive workload (NASA-TLX). Key limitations are that extremely short pattern durations (0.1 s) cause a sharp decline in recognition (dropping to 33% learning accuracy) due to tactile temporal limits (tau and kappa effects), spatial acuity is restricted (1 cm on the hand dorsum, 1.5 cm on the forearm), and performance declines significantly with age (losing 5-9% accuracy per 10 years after a threshold of 45 years old).

Bongers and van der Veer [49] proposed a gestural input model based on the active role of touch in manipulative processes, defining the concept of redundant tactual articulatory feedback. Their approach utilized a computer mouse integrated with a miniature electromagnetic surface transducer resting directly under the user's finger, driven by low-frequency sound signals. They attempted to address the flat sensory limitations and visual attention overload of conventional keyboards, touchscreens, and mice, which do not provide haptic cues of virtual data.. Their results showed that tactile articulatory feedback significantly improves task performance only when the spatial difficulty of the gestural task is high. Key limitations are that for simple tasks, added haptic feedback acts as sensory clutter and actually decreases user efficiency; the Pacinian corpuscle system has a low frequency/pitch discrimination capability of only 20% to 25% compared to the ear's 1%, and early prototypes were heavily dependent on the physical contact pressure of the finger.

Ipakchian Askari et al. [50] proposed a multi-dimensional design space classification to systematize and analyze the development of Affective Haptic Devices (AHDs). Their approach gathered 89 unique AHD prototypes from ACM and Inspec databases, mapping them across communication (revisability, reviewability) and hardware (actuation type, body location, responsivity) characteristics. The system was designed to address severe terminology fragmentation and lack of systematic design parameters, which has historically led to isolated, non-standardized prototypical hardware in mediated social touch (MST) research. Their results showed that vibration remains the dominant AHD actuation method (44.9% of devices), followed by force (30.3%) and warmth (6.7%), while inputs are driven by touch acts (31.5%), general inputs (30.3%), or abstract controls (19.1%). The major limitations identified are that 85.2% of reviewed systems completely lacked real-time, bi-directional haptic responsivity (which is a core property of natural human touch), and only 38.6% of MST devices supported haptic touch with a secondary communication channel like audio or video.

Wang et al. [51] proposed a unified roadmap of haptic rendering approaches and instability parameters for rigid-body interactions. Their approach applied a systematic review methodology, screening 2,059 database records

down to 42 full-text eligibility assessments to deeply evaluate 25 highly technical papers. The system was designed to address the trade-off between stability and realistic transparency in virtual force feedback, isolating the mathematical and physical parameters that trigger unstable vibrations or "push-through" failures. Their results categorized papers into 13 rendering models (such as penalty, constraint, and passivity), proving that passivity-based control (such as TDPA and MBPA) successfully guarantees absolute system stability. Key limitations include that realistic haptic force computation must maintain a strict, stable update rate of 1000 Hz, and passivity-based algorithms are often too conservative (damping force excessively), which reduces transparency and makes virtual rigid walls feel "sticky" or sluggish.

Terenti and Vatavu [38] proposed Wear+Touch, a taxonomy of four interaction techniques (*addition, reinforcement, extension, expansion*) that leverage commodity and custom wearable devices (smartwatches, rings, armbands) to deliver vibrotactile feedback on the user's arm during touch input. Their approach involved developing demonstrative applications (gaming, text entry, eyes-free controls) and evaluating them across two controlled within-subjects experiments with 40 total participants (20 per experiment). The authors sought to solve the absence of native haptic feedback on displays such as large public touchscreens and expand limited vibration design options on mobile devices. In Experiment 1, substituting display vibrations with wearable haptics produced favorable user experience, with the index finger location significantly outperforming the wrist and forearm for Enjoyment ($F(2, 95) = 13.927, p < .001, \eta_p^2 = .23$) and Efficiency ($F(2, 95) = 24.654, p < .001, \eta_p^2 = .34$); in Experiment 2, *reinforcement* and *expansion* significantly improved Enjoyment (medians 4 and 5 vs. 3) and Efficiency (medians 4 and 4 vs. 3) over on-screen vibrations alone ($p < .05$). Limitations include the lack of a unified software architecture across diverse wearables, the absence of multi-user touch-attribution middleware, and evaluation restricted to a young adult sample.

Quinn et al. [17] proposed Shape, a shape-changing haptic navigation interface that communicates non-visual 2D and 3D spatial guidance through active body bending. They evaluated Shape against a 1-DOF Linear Resonant Actuator vibration mode and natural vision using an HTC Vive VR system (tracking at 100 Hz with < 0.1 mm precision), where 10 visually impaired and 10 sighted participants located 60 sequential 3D targets in a 3m radius spherical frustum. The device was designed to solve the inability of conventional non-visual interfaces to communicate precise spatial directions without inducing the distraction, fatigue, or numbness associated with continuous vibrotactile alerts. Visually impaired participants located targets significantly faster and more efficiently using Shape compared to vibration ($p < 0.0001$), achieving angular rotation efficiency comparable to natural vision

in sighted subjects, while spline bend accuracy remained within $\pm 1^\circ$. Limitations of the approach include testing in a static 2×2 m indoor laboratory setting with virtual VR targets rather than real-world mobile walking environments, the absence of an integrated camera perception system, and sequential target presentation rather than handling multi-target cluttered scenes.

Kim et al. [52] proposed a directional vibrotactile feedback interface using wearable ErgoTac devices for real-time ergonomic postural adjustment. They designed three feedback modalities (SPOT repulsive vibration, RAMP intensity scaling, PATTERN sequential vibration) and tested them on 15 subjects across torso and arm joints, subsequently integrating the superior modality (SPOT) with a real-time joint-overload torque optimization framework on 5 subjects performing heavy lifting (4 kg object at 0.2m, 0.5m, and 0.8m). The system was designed to support ergonomic postural correction in industrial tasks where visual or auditory feedback may be distracting or impractical. SPOT significantly outperformed RAMP and PATTERN in overall reaching time (28.35 s vs. 60.24 s for RAMP and 54.83 s for PATTERN; torso reach time 12.70 s vs. 34.24 s and 30.61 s, $p < 0.001$) and System Usability Scale (SUS) score (77.33 vs. 56.83), while the integrated ergonomic lifting test significantly reduced lower-body joint overload torques ($p < 0.05$) with an average SUS score of 79.58. Limitations include reliance on expensive XSENS motion capture hardware, inconsistent torque reductions in upper-limb joints (e.g., shoulder torque reduced by 27.06%-43.74% while elbow torque showed minimal or negative change), and testing restricted to a small cohort ($N = 5$).

Terenti et al. [53] evaluated the user experience of distal arm-level vibrotactile feedback delivered at finger, wrist, and forearm locations during touch interactions. They conducted two exploratory within-subjects experiments: Experiment 1 ($N = 14$) evaluated target selection on HMD-rendered virtual displays versus a VR controller baseline, while Experiment 2 ($N = 14$) tested physical horizontal and vertical touchscreens. Their goal was to solve the lack of empirical understanding regarding how decoupling touch contact points from tactile feedback locations on the arm impacts user experience across virtual and physical display worlds. In VR (Exp 1), distal arm feedback elicited high perceived enjoyment ($M = 3.9, SD = 0.8$), efficiency ($M = 3.6, SD = 0.9$), confidence ($M = 3.9, SD = 0.8$), and integration ($M = 4.2, SD = 0.6$), with low distraction ($M = 2.1$) and a statistically significant preference for finger location over wrist, forearm, and VR controller ($p < .01, \eta_p^2 = .15$); for physical touchscreens (Exp 2), finger feedback was similarly preferred and rated significantly higher for Enjoyment ($Mdn = 5$ vs wrist/forearm, $p < .01$), Efficiency ($Mdn = 5$), and Preference ($p < .001$). Limitations include small sample sizes ($N = 14$ per experiment), testing only extreme ends of the Reality-Virtuality Con-

tinuum (excluding mixed/augmented reality), and relying primarily on subjective Likert measures.

Jouybari et al. [54] proposed the FeetBack system, an untethered torso-worn vibrotactile garment that remaps walking-related foot pressure sensations onto two linear vibrator arrays on the wearer's back using vibrotactile apparent movement (VAM). In a study with 26 analyzed healthy participants (out of 29 recruited), they evaluated motor awareness (MA) and gait adaptation across 120 over-ground walking trials involving 11 temporal delay conditions (0ms to 1500ms in 150ms steps) and catch trials. The system addresses compromised sensory reafferents in individuals with amputations, peripheral neuropathies, or movement disorders (such as Parkinson's), aiming to restore gait awareness and a sense of agency. Participants reported high motor awareness for non-delayed/synchronous trials ($85 \pm 3\%$), which dropped to a minimum at 600ms delay ($22 \pm 6\%$, half-cycle delay) and rebounded for a full gait-cycle delay at 1050–1200ms ($78 \pm 4\%$), fitting a cosine model (38% reduction in residual sum of squares over a linear fit, $p < 0.001$) with a 50% motor awareness threshold of 378 ± 239 ms. Limitations include minimal gait adaptation in healthy subjects (stride-time deviation altered by only $\sim 3\%$), competing tactile cues from natural ground reaction forces that broadened individual threshold variance, and testing restricted to healthy cohorts.

Tchimino et al. [55] proposed EMG feedback via torso vibrotactile stimulation for proportional closed-loop grasping force control of a hand prosthesis. They conducted a 4-session case study over 4 months with a 54-year-old male transhumeral amputee with an osseointegrated implant, using 4 shoulder/torso tactors conveying 5 discrete muscle contraction levels to guide force-matching tasks using EMG from thoracic (pectoralis major) and contralateral forearm muscles. This approach solves the absence of natural sensory feedback and non-intuitive control muscle sources (e.g., atrophic pectoralis major) in high-level upper-limb amputations where residual forearm muscles are missing. The participant achieved an immediate 87% overall force-matching success rate in Session 1; in Session 2, EMG feedback increased grasping force success rate by $\sim 30\%$ compared to no feedback (73% with feedback vs. 42% and 47% without feedback); and by Session 4, performance using the amputated-side pectoralis muscle (85%) matched that achieved with contralateral thoracic and forearm muscles. Limitations include a single-subject case study design, a sharp drop in performance when feedback was removed (indicating continuous feedback dependency rather than forward model formation), and evaluation limited to structured force-matching tasks rather than daily functional activities.

Lim and Yang [56] evaluated a wearable, sensor-based real-time vibrotactile feedback system for postural intervention to reduce excessive trunk flexion exposure

during construction tasks. 14 novice healthy male participants performed simulated construction tasks (lifting-lowering, shoveling, tying rebar) wearing IMU sensors (T6, thigh, shin, wrist) that triggered short (1s) or long (3s + 30s forced break) vibrations when sagittal trunk flexion exceeded 45°, testing three feedback conditions (no feedback, back, wrist). The study aimed to solve high rates of work-related musculoskeletal disorders (MSDs) and lower back pain among construction workers resulting from frequent or prolonged awkward trunk flexion. Vibrotactile feedback significantly reduced trunk flexion angles across 95th, 90th, and 50th percentiles for lifting-lowering (by 13.7%–47.7%; 50th percentile reduced by 38.3% for back and 47.7% for wrist) and shoveling (by 14.5%–35.0%; 50th percentile reduced by 35.0% for back and 33.5% for wrist) ($p < .05$); productivity was maintained for lifting-lowering and rebar tying while shoveling work rate saw a minor 4.5% reduction, with the wrist selected as the most preferred feedback location. Limitations include complete ineffectiveness at altering postures during rebar tying due to inherent task constraints requiring continuous stooping/squatting (triggering alarm fatigue), short task durations (≤ 8 min) in a lab setting with young novice males ($N = 14$), and unexamined long-term retention.

Barontini et al. [57] proposed the CUFF (Clenching Upper-limb Force Feedback) wearable device, capable of delivering normal force (pressure/squeeze), tangential skin stretch, or combined feedback to the arm. Following physical and psychophysical characterizations ($N = 11$) to determine Just Noticeable Differences (JND), 32 able-bodied participants performed object size/softness discrimination tasks (comparing linear vs. non-linear exponential/logarithmic mappings) and a fragile object manipulation task using CUFF alongside the Pisa/IIT Soft-Hand. The device solves the lack of lightweight, wearable haptic displays capable of simultaneously delivering both normal pressure and tangential force/proprioceptive cues to the arm for teleoperation, prosthetics, and guidance without unencumbering the hands. Psychophysical JND was 2.21 N for normal force and 2.70–2.91 mm for tangential displacement; exponential/logarithmic mapping significantly improved discrimination accuracy over linear mapping for size/proprioception (81.45% vs. 68.75%, $p = 0.007$), softness/force (77.54% vs. 64.26%, $p = 0.0129$), and combined stimuli (68.72% vs. 61.73%, $p = 0.0158$); in fragile object handling, CUFF feedback significantly reduced wrong grasps compared to no feedback ($p = 0.0039$). Limitations include testing restricted to able-bodied participants on their dominant arm, directional cues limited to left/right rotations without up/down stimuli, and occasional perceptual overshadowing during simultaneous multi-modal stimuli (stretch + squeeze).

E. Visual Feedback

Visual feedback is a fundamental interaction modality that communicates information through visual cues such as colors, graphical representations, virtual objects, overlays, and video demonstrations. It supports users in understanding spatial relationships, system states, task progress, and movement performance, and has been applied to collision awareness in virtual environments [27], clinical training and target localization [58], rehabilitation and movement correction [59], spatial pointing and alignment [60], physical skill acquisition [61], and the interpretation of ambiguous natural-language commands [62]. These applications demonstrate the potential of visual feedback to improve user awareness, accuracy, learning, and task performance. However, visual clutter, occlusion, limited field of view, cognitive workload, and the need for continuous visual attention can affect effectiveness, making careful selection and presentation of visual cues essential for effective interaction.

Wang et al. [27] proposed a systematic investigation of visually enhanced collision feedback techniques, defining a taxonomy across five levels of informativeness: Coloring, See-through, Gradation, Deformation, and Hybrid. Their approach utilized a virtual reality (VR) environment where a user moved a virtual hand inside a confined-occluded box to touch a target ball, rendering real-time proximity and collision events through different visual shaders. The system was designed to address degraded hand movement and collision awareness in confined-occluded spaces, where physical obstacles block the user's direct line of sight. Their results showed that See-through (translucent) rendering achieved the shortest task completion time ($M = 3.95$ s, $SD = 2.07$ s) compared to other conditions ($p < 0.007$), and elicited the highest subjective visuo-tactile synesthesia rating ($M = 4.39$, $SD = 1.79$, $p < 0.019$), while the Hybrid condition reduced the collision time ratio (CTR) to its lowest ($M = 0.082$, $SD = 0.068$, $p < 0.03$). However, the major limitations of this approach include that the Hybrid condition significantly extended task completion time ($M = 8.11$ s, $SD = 3.95$ s), which was slower than the Baseline ($Z = 4.28$, $p = 0.030$) and caused cognitive overload, whereas the Coloring-only condition resulted in poor subjective collision awareness ($M = 2.33$, $SD = 1.46$) and made users feel highly stressed.

Lalitharatne et al. [58] proposed an emotional and anatomical data-compression feedback interface using a robotic patient avatar (*MorphFace*). Their approach involved translating 60 high-dimensional pressure sensor readings from a physical silicone phantom model into 4 simplified visual facial expressions of pain (Percentage-Weighted Action Units) projected on a virtual face in real time. The system was designed to address the high cognitive workload, low overview efficiency, and poor target localization accuracy of traditional raw haptic 2D

TABLE IV: Summary of Haptic Feedback Studies

Ref.	Haptic Technique / Property	Key Finding / Quantitative Evidence	Contexts / Applications	Limitations
Haghighi et al. [47]	Amplitude, waveform, duration, rhythm, and spatio-temporal vibration patterns.	With 16 participants under cognitive load, amplitude and duration gave the highest accuracy (median 38 and 39/40); mean scores ranged 35.875–37.125.	Ambient wearables, notifications, and multitasking under cognitive load.	Rhythm took longer to identify; female and non-native participants reported higher NASA-TLX stress.
Kourtesis et al. [13]	Electrotactile feedback varying pulse width, frequency, current, and spatial mapping.	Review of 78 studies; Tacttoo achieved 1.15 mm spatial resolution. Only 13% of studies involved clinical patients.	Prostheses, touchscreen overlays, smart devices, and VR typing.	66% were pilot studies with $N \leq 10$; only about one-third combined tactile and kinesthetic cues; automatic calibration was rare.
Jiang et al. [14]	Pressure, vibration, kinesthetic, skin-stretch, and thermal assistive haptics.	Across 132 papers, pressure was most common (79/132), followed by kinesthetic (36/132) and vibration (32/132).	Graphics, navigation, education, training, and work for BLV users.	Kinesthetic feedback offered higher resolution (11% Weber fraction) than skin stretch (35%); hardware and customization remain constraints.
Bouzbib et al. [15]	Haptic systems classified by physicality and degree of actuation.	Compared four encountered-type haptic systems across different interaction scenarios.	VR interaction, medical training, palpation, surgery, and industrial assembly.	High hardware cost; actuated systems introduce safety, robustness, noise, and wind issues.
Xu et al. [16]	Directional vibrotactile patterns with immediate or concatenated feedback.	Four-LRA wristband achieved about 80–90% recognition under load; median accuracy reached 92% at 0.8 s.	Sequential gestures, VR/AR/XR, and wearable shortcut commands.	Very short patterns reduced accuracy to 67% testing and 33% learning; spatial acuity is limited and performance declined after about age 45.
Bongers and van der Veer [49]	Finger-level cutaneous vibration targeting Pacinian receptors.	Sensitivity followed a U-shaped 40–1000 Hz curve, with greatest sensitivity around 200–300 Hz.	Mouse interaction, electronic musical instruments, and audio editing.	Frequency discrimination is only about 20–25% versus about 1% for hearing; redundant haptic cues can reduce effectiveness.
Ipakchian Askari et al. [50]	Vibration, thermal, force, and skin-stretch feedback for mediated touch.	Analysis of 89 affective haptic devices found vibration most common (44.9%), followed by force (30.3%).	Affective communication, mediated social touch, and physiological sharing.	85.2% lacked real-time bidirectional haptics; only 38.6% combined haptics with audio or visual feedback.
Wang et al. [51]	Force-rendering and collision methods including penalty, impulse, constraint, TDPA, and MBPA.	Screened 2,059 records and retained 25 technical papers covering 13 rendering approaches and 12 instability factors.	VR, rigid-body interaction, surgical/dental simulation, and teleoperation.	Stable force rendering requires about 1000 Hz updates; extending models to 3-DOF/6-DOF interaction is complex.
Terenti and Vatavu [38]	Wear+Touch vibrotactile feedback at finger, wrist, or forearm.	Finger feedback improved Enjoyment ($F(2, 95) = 13.927, p < .001, \eta_p^2 = .23$) and Efficiency ($F(2, 95) = 24.654, p < .001, \eta_p^2 = .34$). Reinforcement/expansion also improved both ($p < .05$).	Touchscreens, in-vehicle interaction, text entry, and gaming.	No unified architecture or multi-user attribution; young-adult sample only ($N = 40$).
Quinn et al. [17]	Shape-changing 2-DOF haptic handle for non-visual spatial guidance.	Visually impaired users localized targets faster and more efficiently than with vibration ($p < 0.0001$); spline accuracy remained within $\pm 1^\circ$.	Non-visual navigation, obstacle avoidance, and target localization.	Static indoor VR; no integrated camera perception; sequential single targets only.
Kim et al. [52]	ErgoTac using SPOT, RAMP, and PATTERN vibrotactile guidance.	SPOT reduced reaching time to 28.35 s vs. 60.24 s and 54.83 s ($p < 0.001$), improved velocity ($p < 0.05$), and achieved SUS=77.33. Joint overload also decreased ($p < 0.05$).	Ergonomic posture correction and industrial lifting.	Expensive XSENS tracking; inconsistent upper-limb torque reductions; small optimization sample ($N = 5$).
Terenti et al. [53]	Coin-motor vibration at the finger, wrist, or forearm.	VR ratings were Enjoyment $M = 3.9$, Efficiency $M = 3.6$, Confidence $M = 3.9$, and Integration $M = 4.2$; finger feedback was preferred on physical displays ($p < .001$).	VR displays, tabletops, and physical touchscreens.	Small samples ($N = 14$ per experiment); no mixed/AR conditions; mainly subjective ratings.
Jouybari et al. [54]	FeetBack torso garment using vibrotactile apparent movement from foot pressure.	Motor awareness was $85 \pm 3\%$ at 0 ms, $22 \pm 6\%$ at 600 ms, and $78 \pm 4\%$ at 1200 ms; 50% threshold was 378 ± 239 ms.	Gait perception, motor awareness, agency, and rehabilitation.	Limited gait adaptation; natural foot cues increased variability; healthy participants only ($N = 26$).
Tchimino et al. [55]	Closed-loop EMG feedback through four torso vibrotactile tactors.	Force-matching success was 87% in Session 1; Session 2 reached 73% with feedback vs. 42%/47% without it; pectoralis control reached 85%.	Proportional prosthetic grasp control after high-level amputation.	Single participant ($N = 1$); continuous feedback dependency; structured desktop tasks only.
Lim and Yang [56]	IMU-triggered vibration at wrist or upper back for trunk-flexion correction.	Trunk flexion decreased 13.7–47.7% during lifting and 14.5–35.0% during shoveling ($p < .05$); reduction reached 47.7% at the wrist.	Construction lifting, shoveling, and rebar tying.	Ineffective for rebar tying; possible alarm fatigue; short lab tasks; novice males ($N = 14$); no retention test.
Barontini et al. [57]	CUFF delivering pressure, skin stretch, or combined upper-arm feedback.	JND was 2.21 N for pressure and 2.70–2.91 mm for stretch. Non-linear mappings achieved 81.45%, 77.54%, and 68.72% accuracy; wrong grasps decreased ($p = 0.0039$).	Prosthetics, teleoperation, robotic guidance, and object discrimination.	Able-bodied dominant-arm testing; left/right cues only; combined cues sometimes caused perceptual overshadowing.

color maps during clinical breast palpation training. Their results indicated that the face-mediated approach significantly improved target localization accuracy to a mean score of 83.8% compared to 67.4% for standard visual-tactile maps ($p = 0.0172$), allowing users to locate hidden nodules in an average of 42 s. However, the limitations of their approach include that the time-to-click advantage was not statistically significant compared to the visual-tactile map ($t_{15} = 1.0040, p = 0.3313$), the number of unsuccessful trials exceeding 90 s remained statistically identical ($t_{16} = 1.1663, p = 0.2606$) with an average of 3 unsuccessful trials for the face vs. 2 for the map, and users could not easily discern the exact raw pressure values of individual sensors.

Lee and Song [28] proposed a visual target deflection and closed-loop posture mapping strategy. Their approach utilized markerless RGB-D camera tracking to dynamically deflect or move a virtual target away from the patient's hand in real time if compensatory body motions (such as trunk leaning) were detected. The system was designed to address undesirable compensatory movement patterns (e.g., trunk leaning) that degrade rehabilitation quality and delay motor recovery in stroke survivors. Their results showed that fusing a doubled visual feedback multiplier (Condition #4) successfully forced patients to suppress compensatory motions, yielding the lowest average trunk range of motion (ROM) in stroke survivors compared to baseline. However, the major limitations of this approach are that the deflection strategy significantly increased the patient's subjective rate of perceived exertion ($p < 0.05$ on the Borg rating scale), prolonged task completion times, and extreme visual feedback (Condition #6) significantly extended movement times for the able-bodied control group ($p < 0.05$).

Woodworth and Borst [60] proposed collaborative visual pointing wands and virtual 3D spatial pointers for mirrored third-person perspectives. Their approach involved projecting customized visual geometries (Target Rods, Sphere Cutouts, Pointing Fans, and striped Pointing Wands) from a body avatar inside a third-person virtual mirror environment. The system was designed to address spatial alignment, depth-judgment blind spots, and collaborative pointing errors that occur when pointing at distant targets from a mirrored perspective. Their results showed that aided third-person pointing was statistically significantly more accurate, producing substantially lower azimuth and elevation errors compared to unaided or monoscopic pointing ($p < 0.001$), with Target Rods being highly effective at minimizing horizontal azimuth deviations. However, the limitations of their approach include that Sphere Cutout wands provided significantly worse pointing accuracy in the front region due to occluded shading, Target Rods lacked vertical elevation indicators leading to alignment errors, and large visual pointers (like Pointing Fans) heavily obstructed the field of view and diminished user immersion.

Ke et al. [61] proposed an immediate Visual Feedback (iVF) video replay system. Their approach involved video-recording spinal cord injury (SCI) patients executing physical skills (like ramping, wheelies, and curbing) and presenting the footage in a side-by-side split screen with an expert model for terminal visual evaluation. The system was designed to address usability and learnability limitations of verbal instruction in advanced Wheelchair Skill Training (WSTr), where complex temporal-spatial dynamics of wheel balancing are difficult to explain. Their results showed that the system produced advanced physical skill training outcomes and performance scores that were comparable or statistically superior to traditional visual-verbal sessions guided by physical therapists. However, the major limitations are that the system is highly restricted by terminal feedback limits, providing 0% real-time correction during active maneuvers, and standard camera perspectives are easily occluded by hands or the wheelchair frame, missing performance errors.

Silva-Batista et al. [63] proposed Mobility Rehab (PT + MR), a therapist-assisted visual biofeedback system designed for gait rehabilitation. Their approach involved a pragmatic clinical trial assigning 28 older adults with and without neurological diseases to either 8 sessions (45 minutes each) of physical therapy combined with Mobility Rehab (PT + MR, $n=22$) or PT alone ($n=6$), using 5 wearable Opal inertial sensors on the wrists, feet, and sternum to display real-time gait metrics on a tablet. The system was designed to address the limitations of subjective and inconsistent clinical gait observation by providing objective, real-time metrics to improve patient-perceived mobility (ABC scale), gait speed, and upper/lower body gait mechanics. In terms of results, PT + MR showed statistically significant improvements in comfortable gait speed (effect size $ES = 0.32$, 95% CI: 0.11 to 0.53, mean speed increase of 0.11 m/s), fast gait speed ($ES = 0.30$, 95% CI: 0.07 to 0.52), and comfortable arm swing range-of-motion ($ES = 0.45$, 95% CI: 0.17 to 0.74), whereas neither group demonstrated significant changes in perceived mobility on the ABC scale (PT + MR $ES = 0.12$; PT $ES = 0.48$). The main limitations included a drastically reduced sample size (28 evaluated out of the planned 200 due to the COVID-19 pandemic and clinic acquisition), lack of long-term retention tracking, and reliance on a clinical assistant for rapid sensor placement.

Luciani et al. [24] proposed an Augmented Reality (AR) visual feedback framework for upper-limb robotic neurorehabilitation. Their approach utilized a Microsoft HoloLens 2 headset to project custom 3D holographic trajectory tunnels with real-time hand tracking and color-coded error feedback across three confidence intervals (10 cm, 6.5 cm, and 3 cm), evaluated on 15 healthy participants wearing the AGREE robotic exoskeleton across 4 exercises, alongside usability testing by 12 clinicians. The

system was designed to address the lack of 3D kinematic feedback in upper-limb rehabilitation exoskeletons, which obscures movement execution quality, reduces therapist acceptance, and risks maladaptive plasticity. In terms of results, the AR system significantly reduced global end-effector tracking error from 2.28 ± 0.45 cm (without HoloLens) to 2.06 ± 0.57 cm (C1), 2.06 ± 0.59 cm (C2), and 2.09 ± 0.56 cm (C3), reduced joint space error from $9.58 \pm 0.86^\circ$ to $8.38 \pm 1.20^\circ$ (C3), and achieved a System Usability Scale (SUS) score of 67.7 ± 12.1 along with a Technology Acceptance Model (TAM) Willingness to Use score of 4.38 out of 5. The main limitations included tracking instability when the user's hand was too close or far from the visor, sensitivity to lighting/reflection variations, lack of evaluation on motor-impaired stroke patients, and therapist-reported complexity requiring extra training.

Zhang et al. [23] proposed and evaluated three AR visual instruction designs for manual precise object manipulation. Their approach involved a user study with 18 participants performing 6 Degrees of Freedom (6 DOF) object alignment tasks (speed tasks and accuracy tasks) under three AR instructions: logical deviation baseline ("only-logical"), precise numerical text ("number-based"), and intuitive color mapping ("color-based") using a HoloLens 2 HMD and an ArUco-tracked physical cube. The system was designed to address the uncertainty surrounding whether detailed real-time visualizations of position and orientation deviations improve execution speed, accuracy, perceived ease-of-use, and mental workload in 6 DOF manipulation tasks. In terms of results, no statistically significant differences were found in physical task completion speed or accuracy among the three instructions (all achieved < 2 mm average position deviation and $< 1^\circ$ orientation deviation in accuracy tasks); however, detailed deviation instructions received significantly higher user preference rankings in accuracy tasks (mean ranks 1.61 for number-based and 1.72 for color-based vs 2.67 for only-logical, $p = 0.002$), and color-based feedback yielded significantly higher perceived enjoyment ($p = 0.008$), efficiency ($p = 0.009$), and predictability ($p = 0.005$). The main limitations included a ceiling effect caused by bare-hand physiological tremors during task execution/release, the narrow field of view and color gamut of the HoloLens 2, and reliance on student participants rather than expert industrial assemblers or surgeons.

Lu et al. [64] proposed a home-based guidance training system featuring interactive real-time visual feedback using Microsoft Kinect. Their approach engaged 12 chronic stroke survivors with moderate-to-severe motor impairment in an 800-minute home training program (20–40 min/session, 3 sessions/week over ~ 5.8 weeks) utilizing Kinect V2 3D skeleton tracking to guide 6 core and 11 optional therapist-prescribed exercises. The system was designed to address the lack of real-time

movement feedback and flexible scheduling constraints in home rehabilitation for stroke survivors with severe motor deficits. In terms of results, the intervention produced statistically significant improvements in Fugl-Meyer Upper Limb scores (FMA-UE: 31.00 to 43.58, $p < 0.001$), Lower Limb scores (FMA-LE: 18.08 to 24.08, $p < 0.001$), total score (FMA-Total: 49.08 to 67.67, $p < 0.001$), active range of motion (AROM shoulder flexion increased by 30.3° to 133.98° , reaching the functional ROM upper limit), and reduced muscle spasticity (total MAS decreased from 8.96 to 6.67, $p = 0.002$). The main limitations were a small sample size (12 participants), lack of a randomized control group, absence of long-term retention testing, and failure to reach statistically significant improvements on the Berg Balance Scale (BBS, $p = 0.137$) or achieve normal active ROM levels of healthy older adults.

Patarini et al. [29] proposed a multimodal evaluation model investigating the interaction between visual feedback (FB) types and physiotherapist-patient (Pht-Pt) interaction levels during treadmill robot-assisted gait training (t-RAGT). Their approach assessed 10 individuals with incomplete Spinal Cord Injury (SCI) using eye-tracking (ET) and high-density EEG across 7 Lokomat robotic sessions combining 3 visual FB types (chart, emoticon, game) and 3 Pht-Pt interaction levels (high, medium, low). The system was designed to address the lack of objective understanding regarding how feedback design and therapist interaction impact patient visual attention, cognitive processing, and emotional engagement in a triadic (patient-robot-therapist) setup. In terms of results, the game FB monopolized visual attention on the monitor ($\sim 90\%$ fixations regardless of therapist interaction), whereas chart and emoticon FBs resulted in $\sim 70\%$ monitor fixations with gaze shifting to the therapist (up to 15%) and surroundings (up to 26% when interaction was low); game FB also elicited significantly lower mental demand (38.00 vs 65.00 for chart) and physical demand (66.00 vs 81.00 for chart), while 10-Meter Walk Test speed improved above the MCID threshold (0.06 to 0.16 m/s). The main limitations included a small sample size (10 patients), a brief intervention protocol (7 sessions, yielding non-significant clinical scale changes), and the absence of direct joint torque/force measurements to verify active physical exertion.

Maio et al. [65] proposed a Pervasive Augmented Reality (AR) tool for industrial shop floors designed via Human-Centered Design (HCD). Their approach featured configuration and markerless visualization modules evaluated across two shop-floor user studies: a preliminary study with 26 participants and a comparative study with 10 participants testing three conditions (C1—paper manual, C2—HMD [HoloLens 2], C3—HHD [smartphone]) across sequential and non-sequential picking orders. The system was designed to address picking errors, high cognitive/physical effort, and slow training times caused

by traditional printed paper manuals during complex kit assembly on industrial shop floors. In terms of results, condition C2 (HMD) was 70% faster per component than paper in sequential picking (11 s vs 36 s) and 54% faster in non-sequential picking (13 s vs 28 s), while significantly reducing perceived confusion (median 2 vs 4.5, $p = 0.005$), mental effort (median 2 vs 4.5, $p = 0.011$), physical effort compared to HHD (median 2 vs 3, $p = 0.011$), and achieving highest satisfaction (median 6.5 vs 2.5, $p = 0.001$) and unanimous user preference. The main limitations included HMD hand-gesture recognition failures when operators wore dark/shiny gloves, HHD tracking degradation under factory lighting, HMD battery life constraints, small final sample size (10 participants), and initial worker apprehension regarding performance monitoring or job displacement.

Xue et al. [25] proposed an AR-assisted guidance method specifically designed for assembly and maintenance of avionics equipment in visually occluded areas. Their approach combined a multi-modality anti-occlusion tracking algorithm (fusing RGBD region, depth, and texture features) with automatic assembly step identification and HoloLens 2 3D AR guidance, evaluated in a user study with 30 participants (15 Traditional Guidance [TG] vs 15 AR Guidance [ARG]) across training, assembly, and maintenance tasks. The system was designed to address high operational skill demands, low efficiency, assembly errors, and severe visual blind spots (e.g., occluded bolts and cables in narrow aircraft equipment bays). In terms of results, ARG significantly increased training efficiency by 128.77% (completion time 61.40 s vs 140.47 s, $p < 0.01$), overall assembly efficiency by 29.53% (751.73 s vs 973.73 s, $p < 0.01$), and maintenance efficiency by 27.27% (399.27 s vs 508.13 s, $p < 0.01$), while achieving a ‘GOOD’ total System Usability Scale score (80.83 vs 52.67 ‘OK’) and significantly lower NASA-TLX workload (20.05 vs 34.49). The main limitations included object tracking being limited to target volumes (cannot track small individual parts like loose bolts), algorithm degradation under severe occlusion requiring manual voice overrides, hardware thermal throttling during extended use, and testing restricted to graduate students rather than experienced factory technicians.

Ramalhinho et al. [26] conducted a usability and performance study evaluating video see-through Augmented Reality (AR) overlay in laparoscopic liver surgery. Their approach utilized a phantom surgical setup with 24 participants (12 surgeons, 12 non-surgeons/engineers) localizing 3 virtual liver tumours across 3 image guidance conditions: Baseline A (no guidance), 3D Guidance B (3D model on a separate screen), and AR Overlay C (3D model superimposed on laparoscopic video) using an optically tracked laparoscope and pointer. The system was designed to address the lack of formal quantitative usability and performance evaluation for tracked AR overlays during laparoscopic surgical targeting. In

terms of results, AR Overlay C significantly reduced median localization error for non-peripheral tumours in the surgeon group from 25.8 mm (Baseline A) to 9.2 mm ($p < 0.05$) and in the engineering group from 23.9 mm to 8.2 mm ($p < 0.05$); it also significantly reduced NASA-TLX mental demand (17.5 vs 77.5 for engineering) and surgeon effort (10.0 vs 32.5), with 15 of 24 participants explicitly preferring AR Overlay C. The main limitations included using a rigid phantom setup lacking intra-operative liver tissue deformation, restricting targets to superficial locations due to non-piercing pointers, visual overlay misalignments caused by camera calibration lever-arm effects, minor video streaming latency, and the absence of explicit 3D depth perception cues.

Diller et al. [66] presented SkillAR, a novel HMD-based Augmented Reality feedback technique for motor skill learning. Their approach combined skeleton-based 3D motion capture (Azure Kinect) with an omnipresent visual avatar feedback system on a HoloLens 2 HMD that dynamically adapts, scales, and snaps avatar positions based on user view direction (including top-down and free modes), evaluated in a 32-participant within-subject study against a room-mounted display (RMD) across error identification and pose imitation tasks. The system was designed to address uncomfortable head positions, poor exercise execution, and injury risks caused by room-mounted displays forcing neck twisting during physical exercise. In terms of results, SkillAR enabled significantly higher accuracy in identifying movement corrections compared to RMD (92.71% vs 75.27%, $p = 6 \times 10^{-7}$), while maintaining comparable comparison times (19.01 s vs 18.81 s, $p = 0.92$) and pose imitation joint accuracy (11.14 cm vs 10.71 cm, $p = 0.37$). The main limitations included headset interference during floor-resting or head-contact exercises (e.g., neck stretches), 3D camera tracking inaccuracy for fine/delicate movements, and potential user disorientation when viewing multi-error feedback or non-optimal avatar rotation modes.

F. Multimodal Feedback

Multimodal feedback combines multiple sensory modalities, including visual, auditory, and haptic cues, to provide complementary information and support users during interactive tasks. By distributing information across different sensory channels, multimodal feedback can improve situational awareness, task performance, interaction accuracy, spatial guidance, and user understanding, particularly in complex or attention-demanding environments [30, 39, 31]. It can also reduce reliance on vision alone by providing alternative forms of information through sound and tactile sensations, which is beneficial for eyes-busy, visually demanding, or hands-constrained situations [32, 6]. However, the effectiveness of multimodal feedback depends on appropriate modality selection, temporal synchronization, feedback timing, and information

TABLE V: Summary of Visual Feedback Studies

Ref.	Visual Technique	Key Finding / Quantitative Evidence	Contexts / Applications	Limitations
Wang et al. [27]	See-through rendering, color gradation, deformation, and hybrid collision feedback.	See-through gave the shortest completion time ($M = 3.95$ s, $SD = 2.07$, $p < 0.007$) and highest visuo-tactile rating ($M = 4.39$, $SD = 1.79$). Hybrid achieved the lowest collision-time ratio ($M = 0.082$, $p < 0.03$).	VR assembly, confined-space interaction, and collision awareness.	Hybrid was slowest ($M = 8.11$ s); coloring/deformation caused depth errors, and coloring produced low awareness.
Lalitharatne et al. [58]	MorphFace compressing 60 pressure channels into four facial Action Units.	Target localization improved to 83.8% vs. 67.4% with a visual-tactile map ($p = 0.0172$); 8/16 participants were faster with MorphFace.	Remote clinical examination and prosthetic/medical training.	Localization-time improvement was non-significant ($p = 0.3313$); exact raw pressure values were not visible.
Lee and Song [28]	Markerless visual target deflection and posture mapping.	A doubled feedback multiplier produced the lowest trunk ROM in stroke survivors; 1:1 target-trunk mapping encouraged desired trunk motion.	Upper-limb robotic rehabilitation for stroke survivors.	Stationary-target feedback increased perceived effort; extreme feedback increased movement and exercise time.
Woodworth and Borst [60]	Virtual 3D pointers including Target Rods, Sphere Cutouts, Pointing Fans, and Wands.	Third-person pointing was more accurate than unaided/monoscopic pointing ($p < 0.001$); Target Rods particularly reduced azimuth error.	Collaborative VR and third-person virtual-mirror interaction.	Sphere Cutouts reduced accuracy; Target Rods lacked elevation cues; large pointers could obstruct the scene.
Ke et al. [61]	Immediate visual feedback using side-by-side video replay with an expert model.	Produced comparable or better advanced wheelchair-skill outcomes than traditional therapist-led sessions and exposed spatial-temporal details difficult to describe verbally.	Advanced wheelchair-skill training for SCI patients.	Feedback is terminal rather than real-time; camera views may be narrow or occluded by hands/wheelchair parts.
Silva-Batista et al. [63]	Mobility Rehab using five IMUs to display real-time gait metrics on a tablet.	Comfortable gait speed improved with $ES=0.32$ (95% CI: 0.11–0.53; +0.11 m/s), fast gait speed $ES=0.30$, and arm-swing ROM $ES=0.45$; ABC change was non-significant ($ES=0.12$).	Outpatient gait rehabilitation for older adults with and without neurological disorders.	Only 28/200 planned participants were evaluated; no retention assessment; sensor placement required assistance.
Luciani et al. [24]	HoloLens 2 AR with 3D trajectory tunnels and color-coded movement-error feedback.	End-effector error fell from 2.28 ± 0.45 cm to $2.06\text{--}2.09$ cm; joint-space error fell from $9.58 \pm 0.86^\circ$ to $8.38 \pm 1.20^\circ$; $SUS=67.7 \pm 12.1$.	Upper-limb robotic neurorehabilitation and movement-accuracy training.	Tracking instability and lighting/reflection sensitivity; healthy participants rather than stroke patients.
Zhang et al. [23]	HoloLens 2 AR for 6-DOF alignment using logical, numerical, and color feedback.	All conditions achieved < 2 mm position and $< 1^\circ$ orientation deviation. Numerical/color cues were preferred ($p = 0.002$); color improved enjoyment ($p = 0.008$), efficiency ($p = 0.009$), and predictability ($p = 0.005$).	Precise manual manipulation in industrial and surgical-style tasks.	Hand tremor and HoloLens FOV/color limits; students rather than expert users.
Lu et al. [64]	Kinect V2 skeleton guidance with real-time joint-angle and center-of-mass feedback.	FMA-UE improved $31.00 \rightarrow 43.58$, FMA-LE $18.08 \rightarrow 24.08$, and total FMA $49.08 \rightarrow 67.67$ ($p < 0.001$); shoulder flexion increased 30.3° and MAS fell $8.96 \rightarrow 6.67$ ($p = 0.002$).	Home-based tele-rehabilitation for chronic stroke survivors.	Small single-arm sample ($n = 12$); no retention test; BBS change was non-significant ($p = 0.137$).
Patarini et al. [29]	Visual biofeedback during Lokomat gait training with eye tracking and HD-EEG.	Game feedback attracted $\sim 90\%$ of fixations vs. $\sim 60\text{--}70\%$; mental demand fell from 65 to 38, physical demand from 81 to 66, and 10MWT improved $0.06 \rightarrow 0.16$ m/s.	Robot-assisted gait rehabilitation for incomplete SCI.	Small sample ($n = 10$); seven sessions; non-significant clinical-scale changes; no direct joint-force measures.
Maio et al. [65]	Pervasive AR guidance using HoloLens 2 and smartphone markerless visualization.	HMD guidance was 70% faster for sequential picking (11 vs. 36 s) and 54% faster for non-sequential picking (13 vs. 28 s); confusion, effort, and satisfaction also improved.	Industrial picking, kit assembly, and operator training.	Gesture failures with gloves; lighting-sensitive smartphone tracking; battery limits; small comparative sample ($n = 10$).
Xue et al. [25]	HoloLens 2 AR with RGBD anti-occlusion tracking, step recognition, and 3D guidance.	Training efficiency increased 128.77% (61.40 vs. 140.47 s, $p < 0.01$), assembly 29.53%, and maintenance 27.27%; $SUS=80.83$ vs. 52.67 and NASA-TLX=20.05 vs. 34.49.	Avionics assembly, maintenance, and training in visually occluded spaces.	Limited small-part tracking; degraded under severe occlusion; thermal throttling; graduate-student sample.
Ramalhinho et al. [26]	Video see-through AR overlay of tracked liver and vessel models.	Localization error fell from 25.8 to 9.2 mm for surgeons ($p < 0.05$) and 23.9 to 8.2 mm for engineers; mental demand decreased and 15/24 preferred AR.	Laparoscopic liver surgery and tumour localization using a phantom.	Rigid phantom; superficial targets; overlay misalignment; minor latency; limited depth cues.
Diller et al. [66]	SkillAR HoloLens 2 feedback using Kinect skeleton tracking and adaptive avatars.	Error-identification accuracy was 92.71% vs. 75.27% ($p = 6 \times 10^{-7}$); comparison time (19.01 vs. 18.81 s, $p = 0.92$) and imitation error (11.14 vs. 10.71 cm, $p = 0.37$) were comparable.	Motor-skill learning, exercise, and physiotherapy.	Headset interference; fine-motion tracking errors; possible disorientation with complex feedback.

integration, as excessive or poorly coordinated cues can increase cognitive workload and reduce usability [36, 37, 67]. Therefore, designing effective multimodal feedback requires careful consideration of how different sensory

channels complement one another according to the task context and users' interaction requirements. Triantafyllidis [30] proposed a rigorous empirical evaluation of multimodal human-robot interfaces to determine

how combining sensory stimuli affects human perception and task performance in virtual environments. Their approach utilized a first-person virtual pick-and-place manipulation task in a teleoperation environment, evaluating 25 participants across a full-factorial design of 8 interface conditions combining stereoscopic visual displays (VR HMD), haptic gloves, and event-based auditory alerts (collision alarms and placement dings). The system was designed to address the lack of extensive, empirical comparisons showing how combined haptic and auditory sub-modalities compensate for potentially overloaded visual information during high-precision manual tasks. Their results showed that regardless of task complexity, using a stereoscopic VR HMD increased performance across all measurements by 40% compared to a monocular monitor, while haptic feedback improved outcomes by 10%, and auditory feedback accounted for an approximate 5% performance improvement. Subjectively, combining all three modalities (C8) yielded the lowest mental workload (26.66/100 on the NASA-TLX) and the highest usability (92.50/100 on the SUS), while reducing physical placement distance errors to 1.65–1.76 cm (compared to 19.12 cm on monocular screens). The primary limitations of this approach include the requirement to maintain strict system processing latencies of <15 ms to prevent simulator sickness, and the fact that minor cross-sensory misalignments or latency drift can degrade immersion and trigger an “uncanny valley of haptics.”

Cho et al. [39] proposed SonoHaptics, a non-visual, audio-haptic cursor designed to enable hands-free spatial target interaction. Their approach involved mapping polynomial regression models of color-to-pitch transitions, text-to-audio timbres, and spatialized 3D stereo audio paired with active controller vibrotactile pulses during gaze-based selection. The system was designed to address eye-gaze selection errors, target occlusion, and visual fatigue in display-free smart glasses or highly cluttered XR scenes where users lack a visual cursor. Their results showed that SonoHaptics achieved a competitive target selection error rate of 15.3% (comparable to text-to-speech at 14.4% and outperforming a no-feedback baseline at 18.3%), while significantly reducing target selection times in cluttered scenes from 8.3 seconds (using TTS) down to 5.8 seconds. The primary limitations of their approach are that the abstract audio-tactile mappings are unfamiliar, introducing a high initial cognitive workload ($M = 4.30$ on the NASA-TLX), and selection error rates rise from 5.7% (0 distractors) to 23.4% (3 distractors) as scenes become cluttered.

Guo et al. [31] proposed a bimodal spatial guidance system fusing optical see-through AR overlays with wrist-worn haptic cues. Their approach integrated optical see-through AR target lines with active, wrist-based vibrotactile directional “pull” cues that guide the user’s hand along a 3D path. The system was designed to address depth-perception errors and visual attention overload in

AR-guided physical tasks (such as medical needle insertion) when occlusion, glare, or tracking loss hampers vision. Their results showed that fusing the modalities reduced average depth-alignment error to a low of 5.8 mm ($SD=1.6$) (outperforming AR-only at 8.4 mm and haptic-only at 7.5 mm), dropped target overshooting from 27% down to 9%, and achieved a high SUS score of 88.1/100. The primary limitations of this approach include a speed-accuracy trade-off, where multimodal tasks are slightly slower (9.2 s vs. AR-only 8.0 s) as users cross-check visual lines with physical wrist pulses, and vertical directional cue ambiguity, where Up (82% accuracy) and Down (84%) are less accurate than Left (93%) and Right (94%) due to wrist skin biomechanics.

Ariza et al. [32] proposed an audio-driven haptic synthesis model for hand-tracking interactions in Virtual Reality. Their approach utilized real-world acoustic audio recordings to directly drive the frequency and amplitude of vibrotactile actuators attached to the user’s fingers during VR tasks (such as touching glass or grass). The system was designed to address the flat, static, and unrealistic haptic sensations on standard controllers that do not correspond to the physical or acoustic properties of the virtual materials being manipulated. Their results showed that the combined Auditory+Tactile condition achieved the highest subjective embodiment and preference score (0.984) compared to Auditory-only (0.857) and Tactile-only (0.762), with users overwhelmingly preferring multimodal over visual-only feedback (0.97 score). The primary limitations of this approach are that purely vibrotactile actuators lack physical force-feedback capability, restricting the system’s ability to render real physical weight or stiffness gradients.

Kirmayr et al. [68] proposed a progress-feedback style for agentic, multi-step LLM assistants. Their approach designed the “Planning & Results” (PR) feedback style, which periodically narrates upcoming steps and intermediate results via spoken voice and visual icons at fixed 5-second intervals during long-running tasks. The system was designed to address “ambiguous silence” in agentic systems, where extended processing delays leave users uncertain, increasing cognitive load and damaging trust in safety-critical settings like driving. Their quantitative results across 45 participants in a driving simulator showed that PR feedback significantly improved perceived processing speed ($d_z = 1.01$), enhanced user trust ($d_z = 0.38$), and decreased subjective cognitive task load ($d_z = -0.26$ on the NASA-TLX). The primary limitations are that the study utilized a fixed 5-second update frequency rather than exploring dynamic adaptation, participants were recruited from a single automotive company, and the standardized lane-keeping task does not capture complex, dynamic road variations.

Salous et al. [7] proposed an acoustic-to-vibration translation framework to augment on-screen graphics. Their approach developed an automated translation pipeline

that maps real-world acoustic recordings directly to drive haptic vibration sequences on touchscreens, comparing it against LLM-driven text-to-haptic models. The study addressed the limited realism of touchscreen texture representations by translating acoustic characteristics of physical materials into vibrotactile patterns. Their results showed that real acoustic-to-vibration recordings achieved the highest likeness rankings, while the LLM-driven AI prompt-pattern approach achieved a high realistic score on ceramic textures (median 4, IQR=2.5). The primary limitations are that automated text-to-haptic models suffer from high output variance (IQR=1.63), leading to inconsistent texture representations, and the static database patterns lack real-time customizability.

Chua et al. [69] proposed Soni-Phy, an ambient multi-sensory biofeedback system. Their approach used an Empatica E4 wristband to stream physiological heart rate variability data, modulating the pulsing frequency of a pleasant nature soundscape, visual color gradients, or haptic pulses in real-time. The system was designed to address the high cognitive distraction and work interruption rates associated with active-attention wearable alerts (like alarm buzzers) during workplace stress self-regulation. Their results showed that ambient biofeedback significantly increased interoceptive accuracy to 0.901 compared to a baseline of 0.609 ($p = 0.017$) through an ambient feedback approach. The primary limitations are that the study was conducted on a small pilot cohort ($N = 6$), and user attention and biofeedback processing drop significantly when the cognitive load of the primary task (such as intense typing or mathematical tasks) is saturated.

Mullenbach [6] proposed a tactile touchscreen display interface (TPaD) utilizing active surface-friction modulation. Their approach designed an in-vehicle console interface that pairs visual targeting and slider tasks with active, on-surface vibrotactile click and drag sensations (sine wave transients). The system was designed to address severe driver distraction and elevated eyes-off-road times caused by flat, physically unresponsive in-vehicle touchscreen menus. Their results showed that the visual-haptic (VH) condition reduced median Eyes-Off-Road Time (EORT) by 39% (0.67 s decrease) in targeting and by 19% (0.41 s decrease) in slider tasks, allowing 24% of trials to be completed with 0 seconds eyes-off-road. The primary limitations are that haptic-only feedback is insufficient for continuous in-vehicle operations, as task success rates dropped to 50.0% for sliders and 69.1% for targeting without visual reinforcement.

Clément and Bernardes [34] proposed an empirical evaluation of multimodal feedback in mobile-based digital musical instruments. Their approach conducted a within-subject study with 60 participants comparing auditory-only, auditory-visual, and auditory-visual-haptic conditions during a precise mobile-based musical pitch-tuning task. The system was designed to address high user

hesitation and micro-tuning actions in digital mobile instruments due to the total absence of physical visual and tactile guides. Their results showed that the addition of visual and visual-haptic feedback significantly reduced user hesitation ($p < 0.001$), leading to a much lower percentage of micro-tuning actions. The primary limitations are that the supplementary visual and haptic feedback channels did not improve final pitch-tuning accuracy or overall task completion times.

Calò and De Russis [62] proposed a multimodal visual disambiguation interface to clarify natural language commands. Their approach designed a system that, upon detecting an under-specified vocal input, generates a grid of context-specific images (e.g., representing "coziness" or "relaxation") for user-driven concept selection. The system was designed to address semantic alignment and communication failures caused by ambiguous, subjective, and under-specified natural language instructions in smart environments. Their results showed that image disambiguation outperformed textual lists, achieving a statistically higher intention alignment rating of 3.92 ± 0.97 (out of 5) compared to 3.88 ± 0.78 for text-only options, showing a strong capability to convey abstract concepts. However, the limitations of this approach are that image-based disambiguation yielded a slightly lower overall command-execution efficacy score (3.69 ± 1.01 vs. 4.04 ± 1.10 for direct text options), and it requires high-resolution displays to present the image grids clearly and avoid user visual fatigue.

Sharma et al. [36] proposed psychophysical detection thresholds of haptic-audio asynchrony. Their approach conducted controlled experiments using an active-feedback steering simulator (Sensodrive), triggering motorized steering vibrations and headphones digital audio alerts under varied temporal offsets. The system was designed to address temporal synchronization bottlenecks in lane-departure warnings, determining the exact latency thresholds where users perceive warning cues as disjointed. Their results showed that 50% of drivers detected asynchrony outside the window of -75 ms (audio first) to 110 ms (haptic first), while 20% detected it within -50 ms to 65 ms. The results also indicated an asymmetry in temporal sensitivity, with audio-leading offsets detected more readily than haptic-leading offsets. This finding highlights the importance of controlling relative timing between auditory and haptic warning cues.

Xavier et al. [70] proposed a systematic literature review on pseudo-haptic techniques (PHT) in HCI. Their approach synthesized pseudo-haptic research papers, mapping them by display type, touch interaction, and multimodal fusion methods. The system was designed to address design fragmentation in pseudo-haptic research and a lack of clear guidance on how visual illusions depend on secondary tactile/auditory channels. Their systematic results documented that fusing visual soft surface deformation with physical vibrotactile feedback reduced

physical grasping forces by 24.1% without increasing brain cognitive load (measured via EEG). The primary limitations of current designs are that visual properties can severely interfere with haptic illusions, such as a bright yellow color significantly skewing and degrading the user's perception of simulated stiffness.

Khan et al. [71] proposed a scoping review of gaze and speech integration. Their approach reviewed 103 studies (1991–2025), classifying systems into explicit (deliberate user cues) and implicit (natural behavioral trace) interfaces. The system was designed to address the lack of unified frameworks and temporal alignment protocols for combining gaze and speech. Their scoping results documented that gaze fixations act as a leading indicator, occurring on average ~630 ms before the spoken referent is uttered. The primary limitations highlighted are that explicit systems are highly sensitive to gaze drift and eye tracker noise, while gaze tracking suffers from the 'Midas Touch' problem (accidental activations).

Wakid et al. [67] proposed a systematic review and bibliometric analysis of multimodal HMI design approaches. Their approach analyzed 70 peer-reviewed articles (2019–2024) using the PRISMA framework and Biblioshiny mapping. The system was designed to address integration gaps and cognitive workload management failures across modern HMI platforms. Their results showed that research is heavily concentrated in societal contexts (64.28%), followed by medical (24.29%) and transportation (11.43%). The primary limitations identified in current systems are that slow feedback processing and a response latency of 2–5 seconds in multimodal data fusion remain the single largest barriers to real-world safety deployments.

Su et al. [72] proposed a systematic review of advancements in multimodal human-robot interaction. Their approach screened 359 publications down to 227 empirical HRI studies (2008–2022), compiling gesture, voice, gaze, and bio-signal inputs. The system was designed to address the lack of socially competent robots capable of naturally perceiving user intent and synthesizing multimodal action plans. Their results mapped how deep-learning and transformer architectures fuse continuous human instructions to activate specified robot control modes. The primary limitations of reviewed studies are high computational requirements, system complexity, and a severe lack of real-world longitudinal deployments.

Azofeifa et al. [12] proposed a systematic review and cluster analysis of multimodal HCI. Their approach analyzed 112 key articles, executing t-SNE and hierarchical dendrogram cluster subdivisions to map technologies against domains. The system was designed to address gaps in HCI domain-technology configurations, mapping which modalities (e.g., haptics, VR, tracking) are optimal for specific concept spaces. Their results indicated that VR and haptic interaction were among the most widely represented technologies across medical,

transportation, and industrial application domains. The primary limitations are a lack of standardized evaluation metrics across domains, preventing direct, quantitative performance comparison of HMI architectures.

Hoffmann et al. [73] proposed an empirical comparison of Voice, Touchscreen, and Mid-air gesture modalities. Their approach gathered 429 user suggestions from 13 participants across 11 household/in-vehicle tasks, ranking usability and social acceptance. The system was designed to address the lack of unified design guidelines for smart speaker and automotive graphical user interface control. Their results showed that Voice control received the highest fitting score (6.6/7, SD=0.7) and ease score (6.3/7), closely matching touchscreens (6.3 fitting, 6.4 ease) but significantly outperforming gestures. The primary limitations are that mid-air gestures scored extremely low in social acceptance (2.4 where 1 is best), required high memory retrieval effort, and caused high user fatigue.

Tasaka [37] proposed the Media Adaptive Buffering (MAB) control mechanism. Their approach involved fixing the haptic playout buffer to 10 ms while dynamically adjusting the audio-video buffer from 20 to 150 ms to manage network jitter. The study examined how buffering and network delay influence temporal coordination and perceived quality in haptic-audiovisual communication. Their results identified three critical delay-response thresholds: impact (<25 ms), perception (50 ms), and adaptation (≥ 100 ms). The study examined how buffering and network delay influence temporal coordination and perceived quality in haptic-audiovisual communication.

Holzmeyer et al. [59] proposed a multi-sensory physiotherapeutic guidance strategy. Their approach executed qualitative thematic analysis (Krippendorff's kappa = 0.88) on therapists' hands-on and hands-off movement correction behaviors during exercises. The system was designed to address gaps in mapping physiotherapeutic feedback requirements (modality, timing, frequency) to wearable sensors and haptic shirts. Their results showed that therapists strongly recommended concurrent feedback (70.8%) over terminal-delayed (27.1%), showing that visual/auditory cues are best for starting positions, while haptics guide muscle activation. The primary limitations are that unimodal haptic feedback was barely observed (0.8% of cases), as vibrations lack the high informational bandwidth required to explain posture without visual or voice aids.

Ricci et al. [18] proposed a virtual reality (VR) subway station framework to evaluate a wearable multi-sensory electronic travel aid (ETA) combining a 10-actuator vibrotactile belt for haptic feedback and headset beep sounds for audio feedback. Their approach involved testing 72 healthy participants experiencing simulated glaucoma across four conditions (haptic-only, audio-only, combined, and no feedback) while performing a virtual obstacle-avoidance task. They aimed to solve the lack of empirical evidence, realistic testing environments, and

user-centered multisensory designs in current ETAs to enhance safe mobility and spatial awareness for individuals with visual impairments. Numerically, haptic feedback significantly reduced obstacle collisions ($P = .05$) and head pitch angle entropy ($P = .02$) while increasing task completion time ($P = .03$); audio feedback improved trajectory smoothness ($P = .006$) and mitigated the trajectory length increase caused by haptic feedback ($P = .04$), with 76% (52/72) of participants reporting high engagement and 40% (29/72) preferring combined feedback. Limitations include evaluating only one type of audio beep signal, simulating only glaucoma rather than diverse eye pathologies, lacking real-world collision consequences or physical incentives in VR, using a fixed 10-actuator layout, and evaluating solely healthy subjects with simulated impairment rather than individuals with organic vision loss.

Chappell et al. [74] proposed a non-invasive, closed-loop continuous myoelectric prosthetic hand controller (CLCC) that continuously modulates 3 degrees of freedom (DOFs) paired with a 15-actuator haptic armband rendering multimodal proprioceptive feedback (tangential position, normal displacement, and vibration). Their approach holistically evaluated physical performance and psychological perception in 28 participants without upper limb difference (ULD) across four control groups alongside 10 case-study participants with ULD. They sought to overcome the dexterity limitations, lack of sensory feedback, poor force modulation, and high rejection rates of standard open-loop discrete pattern-recognition prostheses. Numerically, CLCC reduced dual-DOF position matching median absolute error by 54.0% compared to open-loop continuous control ($P < .001$) and median force error by 22.3% ($P < .05$); restored blind-folded object identification accuracy to $54.9\% \pm 4.9\%$ (CLCC) versus 38% for open-loop continuous control ($P < .0083$); significantly increased perceived sensation ratings ($P < .001$); and revealed a significant longitudinal decline in perceived body completeness among participants with ULD ($P < .01$). Limitations include control latency caused by reactive processing and hand mechanics, position accuracy degradation when donning the prosthesis due to added limb weight, lack of direct tactile/slip feedback, and laboratory constraints that limit daily real-life ecological validity.

Pi-Ruano et al. [75] proposed supplementing Variable Message Sign (VMS) road displays with in-vehicle verbal auditory messages during SAE Level 1 partially automated driving. Their approach utilized a driving simulator study with 39 French drivers divided into Visual First and Audio+Visual First groups completing a VMS take-over task and a car-following brake-reaction task while recording driving metrics, EEG spectral power, and self-reported mind-wandering. They sought to solve driver inattention, low vigilance states, and underarousal caused by monotonous environments during partially automated

driving. Numerically, the Audio+Visual modality significantly increased critical VMS message discrimination sensitivity ($d' = 3.80$ vs. 3.54 , $P < .001$) and average response distance (158 m vs. 67 m, $P < .001$); experiencing Audio+Visual messages first produced significantly faster car-following reaction times (1.8 s vs. 2.15 s, $P = .024$) and suppressed the vigilance-decrement-related increase in low alpha EEG power (7.8–9.6 Hz, $P = .024$). Limitations include not isolating verbal from non-verbal auditory cues, an inability to determine the precise time duration of the persistent alertness effect, the absence of a no-message baseline, pre-existing baseline group differences in MAAS mind-wandering tendency, and reliance on a simulated environment rather than real-world traffic testing.

Monsaingeon et al. [35] proposed a Multimodal Interface indicating Limits of Automation (MILA), combining peripheral visual Indicator of Proximity to Limits of Automation (IPLA) cloud displays, earcons, and steering wheel haptic vibrations/stiffness. Their approach involved a 3-week longitudinal simulator study with 40 drivers evaluating MILA against a Visual Basic Interface (VBI) during SAE Level 2 driving across 4 suspension scenarios (curves, traffic jams, erased markings, fog) while tracking driving performance, ocular fixations, mental models, trust, and workload. They aimed to address the out-of-the-loop phenomenon, mode confusion, inappropriate attention allocation, and miscalibrated trust during automation suspensions. Numerically, MILA significantly increased visual fixations on the instrument cluster before suspensions in sharp bends (0.10 vs. 0.02, $P = .001$) and erased markings (0.13 vs. 0.05, $P = .018$); reduced central lane deviation after fog suspensions (0.29 m vs. 0.43 m, $P = .007$); improved mental model accuracy regarding curve limits (6.24/10 vs. 4.30/10, $P = .029$) and traffic jam limits (3.76/10 vs. 2.00/10, $P = .032$); produced greater trust progression across sessions ($U = 122$, $P = .017$); and caused higher subjective workload (21.6/120 vs. 16.2/120, $P = .012$). Limitations include an asymmetry where improved mental model knowledge did not translate into improved take-over driving performance in bends or erased markings, discrete earcons failing to provide continuous anticipation, low scenario realism due to an artificially high event frequency (20 events in 30 mins), repetitive mental model survey prompts, and evaluation limited to 3 weekly sessions rather than daily multi-week usage.

Altukhaim et al. [76] proposed a visual-haptic human-machine interface combining pre-recorded video of a closing/opening hand (visual), pneumatic balloon inflation/deflation (haptic), and finger movement (kinesthetic) during passive hand-grasping. Their approach tested two synchronous feedback conditions—in-phase versus anti-phase—on 23 healthy participants, measuring threat-response reaction times (RT) to an approaching virtual fork to objectively quantify Sense of Ownership (SoO)

over a virtual hand. They sought to solve disembodiment and the loss of SoO over artificial or impaired limbs in neurorehabilitation (such as post-stroke hemiparesis or lower/upper limb amputations). Numerically, 20 out of 23 participants exhibited significantly shorter reaction times under the anti-phase condition compared to in-phase ($P = .000143$ across all subjects), where high pressure from the inflating real balloon paired with visual hand closing created a stronger perceptual SoO (20/23 strongly agreed the virtual hand belonged to their body). Limitations include a focus strictly on passive motion without active motor intention or Sense of Agency (SoA), a small and non-diverse sample size ($n = 23$), lack of condition order counterbalancing, which may introduce training/order biases, the absence of a spatial displacement condition between real and virtual hands, and a lack of quantitative psychophysical pressure measurements. Van Brummelen et al. [33] proposed an audio/haptic smartphone scanning app featuring interactive image segmentation ("MagicTouch"), a 3D bounding-box "Landmark Tracker" Kalman filter algorithm, pitch/speed-modulated music for distance guidance, variable haptic vibrations for framing guidance, and hear-through headphones. Their approach evaluated 50 non-expert participants scanning an outdoor tourist landmark (Agatha Christie Memorial) in a central London field study comparing audio/haptic guidance against a visual-guidance baseline. They aimed to solve the difficulty non-experts face when 3D scanning objects, specifically reducing visual distraction, public social anxiety, safety hazards from walking while looking at screens, and tracking/SLAM drift. Numerically, the audio/haptic app significantly increased user engagement (3.76/5 vs. 3.40/5, $P = .015$), usability (3.95/5 vs. 3.57/5, $P = .049$), reward factor (3.95/5 vs. 3.60/5, $P = .041$), and scan duration (59.2 s vs. 45.0 s, $P = .02$); reduced SLAM drift occurrences from 83% in the pilot to 18% in the main study; and generated significantly more accurate 3D mesh reconstructions (Chamfer Distance 5.49 cm vs. 6.66 cm, $P = .021$; F-score 61.9% vs. 56.6%). Limitations include the audio/haptic interface failing to significantly reduce self-reported stress or physical collision rates (25% audio/haptic vs. 20.8% visual collided at least once), unstudied long-term novelty effects, the exclusion of a combined audio-haptic-visual condition, and testing restricted to a single outdoor landmark location.

V. CROSS-DOMAIN ANALYSIS

The modality-level results show that feedback effectiveness cannot be explained by a single modality hierarchy. Instead, the information being delivered, the sensory and attentional resources used for the primary task, the necessary response time, and the consequences of an interaction error determine the best feedback. Therefore, rather than serving as interchangeable alternatives, auditory, tactile, visual, and multimodal feedback have dis-

tinct functions throughout the reviewed domains. Using the results reported in Section IV, the following cross-domain analysis summarizes recurring patterns across the reviewed application contexts.

A. Driving

Because drivers must constantly monitor the road while engaging with car interfaces and reacting to safety-relevant events, driving imposes particularly severe restrictions on visual attention. Consequently, the evaluated evidence prefers feedback that communicates crucial information without requiring lengthy visual examination. When performing secondary in-car duties, haptic input can lessen dependency on visual interaction. Visual-haptic feedback decreased median eyes-off-road time by 39% during aiming and by 19% during slider interaction in Mullenbach's touchscreen testing; 24% of trials were finished without an off-road glance [6]. However, proper temporal coordination is necessary for non-visual feedback to be beneficial. According to Sharma et al.'s audio-haptic warning study, 50% of users recognized asynchrony outside an interval of roughly -75 ms to 110 ms, indicating that even minor cross-modal timing variations may become perceptually noticeable [36].

The results of the most recent driving tests further show that visual information can be enhanced by another channel rather than necessarily eliminated. According to Pi-Ruano et al., audio-visual traffic communications increased reaction distance from 67 m to 158 m and improved discrimination sensitivity for critical messages ($d' = 3.80$ versus 3.54 [75]). Similarly, the multimodal interface evaluated by Monsaingeon et al. [35] improved several attention, automation-limit knowledge, trust, and lane-location metrics; however, the multimodal condition also increased subjective effort.

During multi-step in-car agent processing, progressive spoken and visual status feedback also reduced subjective workload and increased perceived processing speed and trust [68].

Taken together, the reviewed evidence supports selective use of complementary modalities rather than automatically maximizing the number of feedback channels or removing visual feedback entirely. While visual information can still be helpful for continuous or detailed system states, haptic and auditory cues are helpful when information needs to be noted without diverting sight. These findings highlight low attentional demand, clear allocation of information across channels, and appropriate temporal coordination as important design considerations [6, 36, 75, 35].

B. Cooking and Domestic Tasks

Compared with driving, cooking often involves manual labor and fragmented visual attention, rather than the constant observation of a visual scene crucial for safety.

TABLE VI: Summary of Multimodal Feedback Studies

Ref.	Multimodal Technique	Key Finding / Quantitative Evidence	Contexts / Applications	Limitations
Triantafyllidis [30]	VR, haptic gloves, and auditory alerts.	VR improved performance 40%, haptics 10%, and audio ~5%. Trimodal feedback achieved NASA-TLX=26.66, SUS=92.50, and error=1.65–1.76 cm vs. 19.12 cm.	VR teleoperation and precise manipulation.	Needs < 15 ms latency; cross-modal delay reduces immersion.
Cho et al. [39]	Spatial audio with vibrotactile gaze feedback.	Error was 15.3% vs. 14.4% TTS and 18.3% baseline; selection time fell 8.3→5.8 s.	Hands-free XR target selection.	High initial workload; errors increased with distractors.
Guo et al. [31]	AR overlays with wrist haptic cues.	Depth error=5.8 mm vs. 8.4 mm AR and 7.5 mm haptic; overshoot 27%→9%; SUS=88.1.	AR spatial guidance.	Slower than AR-only; vertical cues less accurate.
Ariza et al. [32]	Audio-driven finger vibrotactile feedback.	Audio+tactile scored 0.984 vs. audio 0.857 and tactile 0.762; multimodal preference=0.97.	VR material interaction.	Cannot reproduce weight or stiffness forces.
Kirmayr et al. [68]	Spoken progress updates with visual icons.	Improved perceived speed ($d_z = 1.01$), trust ($d_z = 0.38$), and workload ($d_z = -0.26$).	LLM assistants during driving ($N = 45$).	Fixed 5-s updates; limited participant/task diversity.
Salous et al. [7]	Acoustic-to-vibration tactile graphics.	Real recordings ranked highest; LLM ceramic feedback median=4 (IQR=2.5).	BLV touchscreen textures.	LLM patterns varied; limited real-time customization.
Chua et al. [69]	Ambient audio, visual, and haptic biofeedback.	Interceptive accuracy increased 0.609→0.901 ($p = 0.017$).	Workplace stress and self-awareness.	Small pilot ($N = 6$); weaker under high workload.
Mullenbach [6]	Visual touchscreen with friction haptics.	Eyes-off-road time fell 39% for targeting and 19% for sliders; 24% of trials needed no glance.	In-vehicle touchscreens.	Haptic-only success dropped to 50.0% and 69.1%.
Clément and Bernardes [34]	Auditory, visual, and haptic feedback.	Visual and visual-haptic cues reduced hesitation and micro-tuning ($p < 0.001$).	Mobile pitch tuning ($N = 60$).	No gain in final accuracy or completion time.
Contreras Perez [62]	Voice with image-based disambiguation.	Alignment was 3.92 ± 0.97 vs. 3.88 ± 0.78 for text; efficacy 3.69 ± 1.01 vs. 4.04 ± 1.10 .	Ambiguous smart-environment commands.	Requires visual display; may increase visual load.
Sharma et al. [36]	Audio alerts with steering-wheel haptics.	50% detected asynchrony outside -75 to 110 ms; 20% within -50 to 65 ms.	Driving warnings.	Audio-first delays were more noticeable.
Xavier et al. [70]	Visual pseudo-haptics with vibration.	Visual deformation plus vibration reduced grasping force by 24.1% without higher EEG workload.	Pseudo-haptic HCI and VR.	Visual properties can distort stiffness perception.
Khan et al. [71]	Gaze and speech integration.	Review of 103 studies found gaze preceded speech referents by ~630 ms.	Gaze-speech interfaces.	Sensitive to gaze drift, noise, and Midas Touch errors.
Wakid et al. [67]	Systematic review of multimodal HMI.	Of 70 studies, 64.28% were societal, 24.29% medical, and 11.43% transportation.	Cross-domain multimodal HMI.	Fusion delays of 2–5 s remain a major barrier.
Su et al. [72]	Gesture, voice, gaze, and biosignals.	Screened 359 papers and synthesized 227 empirical HRI studies.	Multimodal HRI.	High complexity; limited longitudinal deployment.
Azofeifa et al. [12]	Multimodal HCI review and clustering.	Analyzed 112 studies; VR and haptics were widely used across major domains.	Medical, transport, and industrial HCI.	Heterogeneous metrics limit comparison.
Hoffmann et al. [73]	Voice, touchscreen, and gesture comparison.	Voice fitting=6.6/7 and ease=6.3/7; touchscreen scored 6.3 and 6.4.	Household and in-vehicle interaction.	Gestures had low acceptance, memory load, and fatigue.
Tasaka [37]	Adaptive haptic-audio-visual buffering.	Delay thresholds: impact < 25 ms, perception 50 ms, adaptation ≥ 100 ms.	Networked multimodal communication.	Delays > 60 ms reduced agency and performance.
Holzmeyer et al. [59]	Visual, auditory, and haptic physiotherapy cues.	Concurrent feedback was preferred (70.8%) over delayed (27.1%); uni-modal haptics appeared in 0.8%.	Physiotherapy guidance.	Haptics alone lacked enough information for complex posture correction.
Ricci et al. [18]	Vibrotactile belt with audio cues.	Haptics reduced collisions ($P = .05$) and head-pitch variability ($P = .02$) but increased time ($P = .03$); audio improved smoothness ($P = .006$); 76% reported high engagement.	VR navigation with simulated glaucoma ($N = 72$).	Single beep design; simulated impairment; healthy users.
Chappell et al. [74]	Myoelectric control with haptic armband.	Position error fell 54.0%, force error 22.3%, and object ID reached 54.9% ± 4.9% vs. 38% ($P < .0083$).	Multi-DOF prosthetic control.	Latency, prosthesis weight, no slip sensing, lab testing.
Pi-Ruano et al. [75]	In-vehicle audio with visual messages.	Sensitivity $d' = 3.80$ vs. 3.54, response distance 158 vs. 67 m, and RT 1.8 vs. 2.15 s ($P = .024$).	SAE Level 1 driving ($N = 39$).	No no-message baseline; simulator only.
Monsaingeon et al. [35]	Visual automation cues, earcons, and steering haptics.	Fixations increased; fog deviation fell 0.43→0.29 m ($P = .007$), and trust improved ($P = .017$).	SAE Level 2 simulator ($N = 40$).	Knowledge gains did not always improve takeover; only three sessions.
Altukhaim et al. [76]	Visual, pneumatic haptic, and kinesthetic feedback.	Anti-phase feedback shortened threat RT in 20/23 ($P = .000143$); 20/23 reported strong hand ownership.	Virtual-hand ownership ($N = 23$).	Passive task; small sample; no counterbalancing.
Van Brummelen et al. [33]	Audio-haptic smartphone guidance.	Engagement 3.76 vs. 3.40, usability 3.95 vs. 3.57, reward 3.95 vs. 3.60; SLAM drift 83%→18%; Chamfer 6.66→5.49 cm ($P = .021$).	Outdoor 3D scanning ($N = 50$).	No significant stress/collision reduction; one location only.

Hands-free audio instruction is especially important because users must manipulate ingredients and tools while monitoring the procedural state.

The structure and timing of verbal instruction are crucial, as the examined cooking experiments repeatedly demonstrate. An LLM-based conversational cooking as-

sistant could achieve high perceived trustworthiness and low reported task workload, but semantic interpretation errors and repeated dialog remained potential sources of cognitive burden. Hwang et al. [19] found nine recurrent interaction breakdowns in voice-guided cooking and derived eight rewriting goals for converting written instructions into more appropriate spoken guidance. They also identified processing delays of about 2–5 s as a source of conversational disruption.

These results suggest that temporally and language-specific suitable instruction, rather than just hands-free output, is the primary necessity in cooking. Because users can continue to manipulate materials while listening, audio is particularly well-suited for step-by-step and explanatory information. However, when verbal feedback is too lengthy, poorly timed, semantically inaccurate, or unrelated to the job at hand, it loses effectiveness [19, 20, 42]. When persistent geographical or procedural information is required, visual information may supplement voice, but it shouldn't make the user frequently give up on the physical job.

C. Physical Activity, Rehabilitation, and Wearable Movement Guidance

Frequent study of traditional visual interfaces is unpleasant or impossible since physical exercise and rehabilitation require constant body mobility. The findings demonstrate that while each of the three main feedback channels has a specific purpose, they can all be helpful in this area.

Movement can be continuously represented by auditory sonification without the need for visual attention. Pitch direction affected felt lightness, motivation, capability, fatigue, and movement behavior during exercise, according to Singh et al. [8]. Real-time auditory feedback decreased irregular movement length in chronic stroke patients from 39.4% without feedback to 19.3% with feedback, according to Douglass-Kirk et al. [21]. According to other auditory studies, limb motion and gait can be translated into ecological or parameter-based sounds, and the choice of sound representation affects the user experience [43, 45, 46]

Since cues can be linked to specific body regions or directions, haptic feedback offers an additional benefit. In the wearable gesture study by Xu et al. [16], immediate vibrotactile feedback decreased sequential interaction time from 11.58 s to 8.56 s. In the study by Kim et al. [52], directional vibrotactile guiding also helped with optimal posture adjustment, with the SPOT condition providing lower reaching times than the RAMP and PATTERN alternatives. The placement of wearables is a significant aspect in and of itself: Depending on whether feedback was given close to the index finger, wrist, or forearm, Wear+Touch demonstrated significant differences in efficiency and satisfaction [38].

When users are able to comfortably focus their visual attention on movement correction, visual input is still helpful. Improvements in gait measures, upper-limb rehabilitation, home training, and AR-supported motor learning were reported in the reviewed research [63, 24, 64, 66]. Stronger or complex visual feedback, however, wasn't always superior. While Patarini et al. demonstrated that various visual-feedback designs produced distinct patterns of attention and workload during robot-assisted gait training [29], Lee and Song discovered that stronger visual manipulation could increase perceived effort and movement time [28].

As a result, the cross-study pattern is functional rather than modality-specific: visual feedback provides accurate movement-state representation, haptic feedback provides localized or directional correction, and audio provides continuous or high-level movement information. It is most appropriate to combine or alternate between various channels when each offers a unique type of information.

D. Accessibility and Non-Visual Interaction

Due to the possibility of faulty, incorrect, or absent visual information, accessibility alters the function of multi-modal feedback. In certain situations, tactile and auditory feedback can take the place of secondary supplements as major information channels.

The Vid2Coach study illustrates the advantages and disadvantages of providing blind and low-vision users with aural help. Although feedback ratings did not differ significantly from the baseline and the system remained sensitive to camera interference, ambient noise, speech recognition errors, and intent-classification failures, voice-guided task support made it possible to interact with instructional material [40]. Pressure, vibration, kinesthetic, and similar kinds of haptic feedback have been applied to graphical understanding, navigation, education, training, and job tasks for BLV users, according to Jiang et al.'s larger haptic review [14].

The value of spatial haptic guiding is also supported by direct empirical evidence. Quinn et al. discovered that when utilizing the shape-changing interface instead of vibration alone ($p < 0.0001$), visually impaired individuals found targets far more quickly and effectively [17]. Evidence from multisensory navigation further demonstrates that haptic and auditory cues can affect various aspects of mobility: in the study by Ricci et al., haptic feedback decreased obstacle collisions while auditory feedback enhanced trajectory smoothness, and 40% of participants preferred the combined condition [18]. In order to represent visual textures using haptic information, Salous et al. also investigated acoustic-to-vibration mappings [7]. Therefore, complementing non-visual input is supported by the accessibility data, but it does not support the claim that a single non-visual channel will be equally effective for every user or task. Individual sensitivity, device configuration, calibration, information density, and

environmental reliability are consistently identified as significant limits in the findings [13, 14, 17].

E. Walking and Mobile Interaction

Users need to divide their focus between the interface and their surroundings when walking and interacting on mobile devices. Thus, feedback that minimizes ongoing screen inspection is very beneficial. [22]. showed that walking direction in virtual reality may be influenced by spatialized auditory step feedback, resulting in a notable translational redirection of about 1.7 m in the audio-only condition. Although the scan duration was longer and the interface did not significantly lower collision rates, the field study by Van Brummelen et al. offers direct evidence from an outdoor mobile task: audio-haptic guidance during landmark scanning improved engagement, usability, and 3D reconstruction quality compared to visual guidance [33]. In a similar vein, Ricci et al.'s multisensory navigation study revealed that tactile and auditory feedback influenced various mobility results [18].

These results show that lowering visual dependency does not always result in better outcomes, but they also support the idea of low-attention mobile feedback. While haptic feedback can give directional or localized clues, audio may express environmental or cognitive states. When these responsibilities are separate and the additional signals do not interfere with awareness of the physical environment, their combination is most beneficial [22, 18, 33].

F. Surgery, Precision, and High-Stakes Tasks

As mistakes, delays, and unclear indications may have serious consequences, feedback design must be stricter in high-stakes and precision jobs. The findings offer concrete proof from industrial maintenance, AR-guided precise interaction, teleoperation, and surgical targeting. According to [26], AR overlay guidance decreased numerous subjective workload measures and the median localization error for non-peripheral targets in the surgeon group from 25.8 mm without guidance to 9.2 mm. In the same way, AR guidance for avionics assembly and maintenance reduced reported workload while increasing training, assembly, and maintenance efficiency [25]. Additional advantages of distributing information across channels are shown by multimodal evidence. The full visual-haptic-auditory condition in teleoperation produced high usability, low reported workload, and significantly lower placement error than the monocular-monitor condition, according to [30]. Similarly, combining AR and wrist haptics reduced overshoot from 27% to 9% and depth-alignment error from 8.4 mm to 5.8 mm, although the multimodal condition was slower than AR alone [31]. Therefore, it is not typical for high-stakes assignments to call for the maximum number of modalities. Rather, they need highly understandable data with consistent timeliness and a distinct purpose for every channel. When

visual attention is diverted, haptic or auditory information can supplement detailed visual or augmented reality information. Rather than being introduced by default, extra channels must be justified by measurable task needs, as demonstrated by the observed speed-accuracy trade-offs and latency limitations [26, 30, 31].

G. Cross-Domain Patterns

The comparison of these areas reveals a number of recurrent patterns.

First, when information is immediate, spatial, directional, or connected to a particular body region, haptic feedback is most reliably supported. Wearable gesture interaction, ergonomic guidance, non-visual navigation, and precision interaction all exhibit this pattern [16, 52, 17, 31].

Second, when information is sequential, meaningful, explanatory, or consistently connected to movement, auditory feedback is especially helpful. While rehabilitation studies show that sonification can represent a continuous movement state, voice-guided cooking highlights the importance of spoken procedural information [19, 20, 46].

Third, when users are able to use their visual attention properly, visual feedback is useful for information that is spatially accurate, detailed, and durable. Precise manipulation, rehabilitation, industrial guiding, and surgical targeting have all benefited from AR and visual-feedback research [23, 24, 25, 26]. Nevertheless, the visual results also demonstrate that an increase in visual complexity can lead to an increase in workload, occlusion, completion time, or attentional demand [27, 28, 29].

Fourth, when various sources provide complementary information, multimodal feedback works best. Combining channels can improve certain outcomes, according to the AR-haptic study by [31], the auditory-tactile VR study by [32], the multisensory navigation study by [18], and the audio-haptic mobile study by [33]. However, these studies also reveal trade-offs in time, workload, learning, or other outcomes.

Lastly, timing and synchronization are recurring factors that determine feedback quality independently across domains. Delayed conversational processing disrupted cooking interaction [20]; immediate haptic feedback increased interaction efficiency [16]; and audio-haptic temporal mismatch was directly observable in driving [36]. Therefore, rather than a universal modality ranking, the cross-domain evidence favors a task-dependent model of feedback selection.

Table VIII summarizes selected quantitative findings from representative studies across the reviewed application domains. Because the studies used different tasks, participant populations, baselines, outcome measures, and experimental conditions, the reported values are presented for within-study interpretation and should not be treated as directly comparable effect estimates across studies or modalities.

TABLE VII: Cross-Domain Summary of Evidence-Supported Feedback Patterns

Domain	Evidence-Supported Feedback Pattern	Why It Is Effective	Representative Evidence	Evidence Assessment
Driving	Haptic + auditory; selective audio-visual / multimodal	Reduces visual dependence while supporting warnings and system-state information. Benefits depend on complementarity, workload, and temporal alignment.	Visual-haptic feedback reduced eyes-off-road time by 39% for targeting and 19% for sliders [6]. Audio-haptic asynchrony was detectable outside about -75 to 110 ms [36]. Audio-visual messages increased sensitivity ($d' = 3.80$ vs. 3.54) and response distance (158 m vs. 67 m) [75].	Strong: multiple direct studies support non-visual and complementary feedback; synchronization and workload remain constraints.
Cooking / Domestic Tasks	Context-aware auditory / voice	Supports hands-free sequential guidance while users manipulate tools and ingredients. Effectiveness depends on concise, timely, task-aware speech.	Hwang et al. identified 9 interaction breakdowns and 8 rewriting goals [19]. Context-aware voice reduced communication breakdowns, although 2–5 s delays disrupted interaction [20]. LLM-based assistance showed high perceived trustworthiness but semantic and dialogue-load limitations [42].	Moderate: several direct studies support voice guidance, but samples are small and interaction failures remain important.
Physical Activity / Rehabilitation	Auditory sonification + localized haptics; visual when attention permits	Audio supports continuous movement information, haptics provide localized or directional correction, and visual feedback supports detailed movement guidance when gaze is available.	Auditory feedback reduced abnormal movement from 39.4% to 19.3% [21]. Immediate haptic feedback reduced interaction time from 11.58 s to 8.56 s [16]. AR reduced tracking error from 2.28 ± 0.45 cm to 2.06–2.09 cm [24].	Strong: multiple studies support complementary roles for audio, haptic, and visual feedback.
Accessibility / BLV	Haptic + auditory non-visual feedback	Haptics support spatial/directional information, while audio provides semantic or procedural guidance when vision is unavailable or unreliable.	Shape-changing haptics improved target localization over vibration ($p < 0.0001$) [17]. Haptic applications span navigation, graphics, education, training, and work [14]. In multisensory navigation, haptics reduced collisions while audio improved trajectory smoothness [18].	Strong: direct studies and accessibility reviews support complementary non-visual feedback, with personalization and hardware limitations.
Walking / Mobile Interaction	Directional haptic + auditory guidance	Reduces screen dependence while preserving environmental awareness. Audio conveys semantic/environmental information and haptics provide spatial or framing cues.	Auditory step feedback produced about 1.7 m of translational redirection [22]. Outdoor audio-haptic guidance improved engagement, usability, and 3D reconstruction quality, but increased scan time and did not significantly reduce collisions [33].	Moderate: direct evidence exists, but few studies examine long-term real-world mobile use.
Surgery / High-Stakes Precision Tasks	Visual/AR + selective haptic or auditory support	AR supports detailed spatial precision, while complementary non-visual feedback can assist when it conveys distinct information without adding delay or conflict.	AR reduced surgeons' median localization error from 25.8 mm to 9.2 mm [26]. Multimodal teleoperation reduced placement/depth error to about 1.65–1.76 cm versus 19.12 cm with a monocular monitor [30]. AR+haptics reduced depth error from 8.4 mm to 5.8 mm and overshoot from 27% to 9% [31].	Moderate: direct precision and surgical evidence exists, but broader evidence remains partly simulated, AR-based, or teleoperated.

VI. DESIGN FACTORS AND SYNTHESIS

The cross-domain research shows that the choice of modality is not the only factor that affects feedback effectiveness; additional factors include how a feedback signal is timed, encoded, positioned, coupled with other channels, and matched to the user's resources. The modality-level results consist of the following elements.

A. Timing and Immediacy

In order to maintain a clear relationship between the two, feedback should be given in close relation to the relevant action or system occurrence. When compared

to concatenated feedback, immediate haptic feedback in sequential gestural interaction decreased interaction time from 11.58 s to 8.56 s [16]. On the other hand, delays of roughly two to five seconds interfered with the flow of discourse in context-aware cooking interactions [20]. The information being delivered also affects the timing requirements. Periodic intermediate progress feedback throughout longer-running processes may improve perceived system responsiveness, as [68] showed. Therefore, the goal should be to deliver feedback at a time that is meaningful for the corresponding user action, task state, or system event rather than simply minimizing latency in

TABLE VIII: Selected Quantitative Findings from the Reviewed Studies

Domain	Outcome Measure	Reported Finding	Ref.	Modality
Driving	Audio-haptic temporal synchrony	50% of users detected asynchrony outside approximately -75 ms (audio first) to 110 ms (haptic first), indicating sensitivity to relatively small cross-modal timing offsets.	Sharma et al. [36]	Audio + Haptic
Driving	Traffic-message discrimination and response	Audio-visual messages increased discrimination sensitivity ($d' = 3.80$ vs. 3.54 , $p < 0.001$) and response distance (158 m vs. 67 m, $p < 0.001$). Car-following reaction time was 1.8 s versus 2.15 s when Audio+Visual was experienced first ($p = 0.024$).	Pi-Ruano et al. [75]	Audio + Visual
Cooking / Domestic	Conversational-assistant trustworthiness	The LLM-based cooking assistant achieved an overall trustworthiness score of $4.11/5$ on the Voice Usability Scale, while reported task workload remained low.	Chan et al. [42]	Audio / Voice
Cooking / Domestic	Processing latency during context-aware guidance	Context-aware proactive voice interaction reduced communication breakdowns and dialogue-repair loops, but multimodal integration introduced approximately $2-5$ s of response delay.	Jaber et al. [20]	Audio / Voice
Physical Activity / Rehabilitation	Abnormal compensatory movement duration	Real-time auditory feedback reduced abnormal movement duration from 39.4% without feedback to 19.3% with feedback ($F(1, 18) = 9.424$, $p = 0.007$, partial $\eta_p^2 = 0.344$).	Douglass-Kirk et al. [21]	Audio
Physical Activity / Wearables	Sequential interaction time and recognition	Immediate haptic feedback reduced interaction time from 11.58 s to 8.56 s at 0.3 s pattern duration. Durations of $0.3-0.5$ s achieved median recognition accuracy of approximately $81-89\%$.	Xu et al. [16]	Haptic
Physical Activity / Ergonomics	Trunk-flexion reduction	Real-time vibrotactile feedback reduced trunk-flexion exposure by $13.7-47.7\%$ during lifting/lowering and by $14.5-35.0\%$ during shoveling ($p < 0.05$). Shoveling productivity decreased by 4.5% .	Lim and Yang [56]	Haptic
Accessibility / BLV	Non-visual spatial target localization	Visually impaired participants located targets significantly faster and more efficiently using shape-changing haptic guidance than vibration ($p < 0.0001$); spline-bend accuracy remained within approximately $\pm 1^\circ$.	Quinn et al. [17]	Haptic
Accessibility / Prosthetics	Position matching, force control, and object identification	Closed-loop multimodal feedback reduced dual-DOF position matching error by 54.0% ($p < 0.001$) and force error by 22.3% ($p < 0.05$). Blindfolded object identification reached $54.9\% \pm 4.9\%$ versus 38% with open-loop continuous control ($p < 0.0083$).	Chappell et al. [74]	Multimodal Haptic
Walking / Mobile	Auditory walking redirection	Auditory step feedback produced a significant translational redirection of approximately 1.7 m (median 205 cm vs. 10 cm baseline, $p < 0.001$). Combined auditory and eye-blink visual redirection increased the effect to approximately 2.5 m (median 235 cm, $p < 0.001$).	Weller et al. [22]	Audio / Audio + Visual
Walking / Mobile	3D scanning accuracy and task time	Audio-haptic guidance improved reconstruction quality (Chamfer distance 5.49 cm vs. 6.66 cm, $p = 0.021$; F-score 61.9% vs. 56.6%), but increased scan duration from 45.0 s to 59.2 s ($p = 0.02$).	Van Brummelen et al. [33]	Audio + Haptic
Surgery / High-Stakes Precision	Surgical target-localization error	For surgeons, AR overlay reduced median localization error for non-peripheral liver targets from 25.8 mm without guidance to 9.2 mm ($p < 0.05$).	Ramalhinho et al. [26]	Visual / AR
High-Stakes / Precision Guidance	Depth-alignment error and target overshoot	Combined AR and wrist-haptic guidance reduced depth-alignment error from 8.4 mm (AR only) to 5.8 mm and reduced target overshoot from 27% to 9% . The multimodal condition achieved SUS = $88.1/100$ but was slower than AR only (9.2 s vs. 8.0 s).	Guo et al. [31]	Visual + Haptic

all situations.

B. Location and Spatial Mapping of Feedback

For haptic feedback, the body location at which a cue is delivered can be as important as the cue itself. When vibrotactile feedback was displayed at various points on the hand and arm, Wear+Touch demonstrated notable variations in satisfaction and perceived effectiveness [38]. Additionally, directional wearable haptics showed that users may be guided toward preferred postures by physical placement and mapping [52].

In non-visual instruction, the significance of spatial mapping is especially noticeable. For visually impaired individuals, [17] discovered that shape-changing haptic guidance enabled quicker and more effective spatial target location than vibration. These findings suggest that rather than being chosen just for device convenience, haptic location should have task-relevant relevance.

C. Signal Encoding and Discriminability

The ability of users to consistently identify the meaning of a feedback signal depends on its physical or perceptual

encoding. Among the haptic characteristics studied under cognitive load, Haghighi et al. discovered that amplitude and duration gave the best recognition performance [47]. Similarly, Xu et al. [16] showed that longer vibrotactile patterns supported higher recognition accuracy, whereas very short patterns substantially reduced recognition performance.

For auditory feedback, encoding is equally important. While [46] discovered that participants' experiences varied across sonification types even when all were mapped to the same movement task, [45] reported differences in user preference across sonification configurations, with lower-pitched feedback receiving higher pleasantness ratings. Therefore, choosing a modality is not enough; the signal also needs to be perceptually distinct, understandable, and task-appropriate.

D. Redundancy and Complementarity

The idea that increasing the number of modalities naturally enhances interaction is not supported by the results. Rather, when each channel plays a distinct informative role, multimodal benefits are greatest.

Although multimodal trials were slower, [31] showed that combining AR guidance with directional wrist haptics reduced depth error and overshoot in comparison to AR-only interaction. [32] also found higher embodiment and preference for combined auditory-tactile feedback than for either channel alone. Ricci et al. discovered that haptic and auditory input affected many mobility metrics in navigation, demonstrating how modalities can offer various types of assistance [18].

Multimodal feedback can also introduce additional interaction costs. Adding visual and haptic feedback decreased hesitation, but it had no effect on task-completion time or ultimate tuning accuracy, according to Clement and Bernardes [34]. Therefore, deliberate complementarity is the relevant principle. Redundant presentation is most justified when failure to notice critical information could have serious consequences; otherwise, different channels should convey complementary, task-relevant information.

E. Synchronization and Latency

The temporal relationship between multiple channels representing the same event is an important design variable. Sharma et al. demonstrated that during driving, users were able to identify audio-haptic temporal misalignment within a reasonably small range [36]. Tasaka reported diminished agency with greater buffering delays and similarly found different delay-response ranges in haptic, auditory, and visual communication [37].

In multimodal teleoperation, where cross-sensory misalignment may lower concentration and performance, low latency was also found to be crucial [30]. These results suggest that synchronized modalities should not be viewed as independently timed outputs, but rather as components of the same feedback event.

F. Cognitive Workload and Information Density

Whether feedback supports or interferes with the primary task depends partly on the amount and complexity of information presented. The Hybrid condition produced a lower collision-time ratio in Wang et al.'s visual collision-feedback study, but it required significantly longer task time than the quicker See-through condition [27]. Sono-Haptics demonstrated growing inaccuracy as visual clutter grew in gaze-based XR interaction [39]. Additionally, evidence from multimodal driving indicates that increased perceived workload may coexist with better automation status information [35].

On the other hand, proper representation might lessen workload. While progressive audio-visual feedback during agent processing decreased subjective workload [68], game-based visual feedback in robot-assisted gait training produced lower reported mental demand than chart feedback [29]. Therefore, the appropriate design goal is to give the bare minimum of information needed for trustworthy action and understanding rather than to maximize the quantity of feedback.

G. Context and Available User Resources

The relationship between modality and the resources previously used by the main task is the last recurrent factor. Physical activity necessitates input that is compatible with movement; driving restricts the availability of visual attention; cooking often restricts manual contact; and accessibility may completely eliminate vision as a dependable information channel.

The synthesized evidence suggests three central questions for feedback selection: what information must be communicated, which sensory or motor resources are available, and how quickly the user must respond. Haptic feedback is particularly suitable for localized, spatial, directional, or immediate information; auditory feedback for semantic, sequential, or continuous movement-related information; visual feedback for detailed and persistent state information; and multimodal feedback when complementary information can be distributed meaningfully across channels.[19, 16, 26, 31].

Rather than choosing a feedback modality based on a specific modality hierarchy, the examined research suggests that the modality should be chosen based on the kind of information that has to be transmitted. Spatial, directional, localized, and instantaneous information was most frequently linked to haptic feedback; semantic, sequential, explanatory, and continuous movement-related information was linked to auditory feedback; and detailed, persistent, and spatially accurate representations were linked to visual feedback. When other channels delivered complementary information or when access to another sensory resource was limited by the primary activity, multimodal feedback showed the clearest benefits. The mapping in Table IX should be interpreted as an evidence-informed design heuristic rather than a univer-

TABLE IX: Evidence-Informed Mapping Between Information Type and Feedback Modality

Information Type	Recommended Modality	Design Interpretation	Representative Evidence
Spatial, directional, or body correction	Haptic / Vibrotactile	Use spatially mapped tactile cues for localization, direction, posture, or immediate physical correction without requiring continuous visual attention.	[17, 52, 56, 38]
Sequential or semantic instruction	Audio / Voice	Use concise spoken feedback for step-by-step, explanatory, or context-sensitive guidance, especially when the user's hands are occupied.	[19, 20, 42]
Continuous movement or body state	Auditory Sonification	Map movement changes to continuous sound when users need ongoing feedback while remaining visually engaged with the physical task.	[8, 21, 45, 46]
Detailed or persistent state	Visual / AR	Use visual feedback for information-rich, persistent, or geometric content when sufficient visual attention is available.	[24, 23, 25]
Precise alignment or manipulation	Visual / AR + Haptic	Use visual/AR feedback for detailed spatial state and add haptics when directional or depth-related correction provides additional value.	[26, 31]
Non-visual navigation and accessibility	Haptic + Audio	Use haptics for spatial/directional guidance and audio for semantic or environmental information when visual access is limited.	[17, 18, 33, 14]
System progress / processing status	Audio + Visual	Use concise intermediate status cues during longer-running operations to reduce uncertainty without creating excessive interruptions.	[68]
Immediate, complex, or safety-critical information	Complementary Multimodal	Use multiple channels when information is urgent or complex, but assign each modality a clear role and minimize latency, unnecessary redundancy, and cross-modal conflict.	[36, 75, 35, 30]

sal modality hierarchy. The appropriate modality may change according to task demands, environmental conditions, available sensory and motor resources, accessibility requirements, time pressure, and the consequences of an interaction error. Furthermore, the effectiveness of a modality depends on its implementation, including timing, encoding, body location, information density, and synchronization with other feedback channels.

VII. EVIDENCE-INFORMED DESIGN GUIDELINES

The following guidelines translate recurrent design characteristics into useful recommendations for multimodal feedback systems in hands-busy and eyes-busy interaction based on the findings collected from the reviewed studies. Rather than widely accepted design standards, these proposals are evidence-based guidelines. The task, participant population, feedback implementation, and evaluation methodology of the analyzed research vary significantly. As a result, the recommendations should be evaluated within the specific context in which they are applied.

A. Audio Feedback Guidelines

Auditory feedback is most suitable when information is semantic, sequential, explanatory, or continuously related to movement. Voice-based interaction is especially useful when users' hands are busy with physical tasks, as illustrated in voice-guided cooking and procedural assistance [19, 20]. Additionally, continuous sonification can communicate movement state without requiring users to look at a display [21, 46, 45].

Designers should:

- When users are unable to easily control or examine an interface, use speech to provide sequential, semantic, and explanatory instructions;
- Keep spoken instructions short and organized to reduce the auditory working-memory demands connected with long verbal instructions [19];
- Instead of giving instructions in a fixed sequence, synchronize verbal prompts with the user's current task status [20];
- Use non-speech sounds or sonification when information can be communicated through ongoing changes in movement, direction, state, or intensity [8, 21, 45];
- Select sound mappings that are understandable and acceptable to users, because sound design can affect preference and task experience [43, 46, 45]; and
- Ensure robust handling of environmental noise, speech-recognition errors, semantic misunderstanding, and response failures in voice-based systems [40, 42].

Therefore, the evidence supports auditory feedback as a channel whose design should match the semantic complexity, timing, and continuity of the information being transmitted, rather than simply as a substitute for visual output.

B. Haptic Feedback Guidelines

Haptic feedback is especially appropriate for immediate, localized, directional, spatial, and body-related information. The results demonstrate that the presence of vibration or force feedback is not the only factor that affects haptic effectiveness; other factors include feedback du-

ration, encoding, body location, and physical mapping. [16, 38, 52].

Designers should:

- use haptic feedback for immediate, directional, spatial, or localized information when continuous visual inspection is undesirable [16, 17];
- Place tactile feedback at body locations that are relevant to the interaction and perceptually effective. Wear+Touch showed significant differences in user experience across finger, wrist, and forearm feedback locations [38];
- Evaluate feedback location empirically rather than assuming that proximal and distal stimulation sites are perceptually equivalent [53];
- Use distinct temporal and spatial patterns. Sequential haptic patterns of 0.3–0.5 s achieved substantially higher recognition than extremely short patterns in the study by Xu et al. [16];
- Match the physical task with the haptic representation. While real-time wearable vibration decreased trunk-flexion exposure during construction tasks [56], directional vibrotactile feedback supported optimum posture guidance [52];
- Avoid excessive or repetitive vibration when it could lead to alarm fatigue, habituation, or distraction [56];
- Provide standardized or adjustable parameters when tactile sense varies across users or stimulation conditions [13, 54].

The evidence demonstrates the importance of haptic methods. For example, the CUFF results show that force and skin-stretch mappings can support discrimination and manipulation beyond simple vibration [57], while shape-changing guidance allowed visually impaired participants to locate targets much faster and more efficiently than conventional vibration [17]. Therefore, rather than assuming all tactile input is equal, designers should choose the haptic mechanism based on the information requirement.

C. Visual Feedback Guidelines

Visual feedback remains useful when information needs to be sent in a detailed, persistent, or spatially accurate manner and when it is safe to allocate enough visual attention. The results show the benefits of visual and augmented reality feedback in industrial guidance, precise manipulation, rehabilitation, and surgical targeting [24, 23, 25, 26].

Designers should:

- Use visual feedback when users require persistent spatial, kinematic, procedural, or system-state information that would be difficult to communicate efficiently through transient auditory or haptic cues;
- When exact localization is needed, spatially align AR feedback with the movement or physical object. AR overlay substantially decreased target-localization error in the surgical study by Ramalhinho et al. [26];

- Use real-time visual representation when continuous movement correction is needed. AR kinematic feedback reduced upper-limb tracking error in the rehabilitation system evaluated by Luciani et al. [24];
- Avoid unnecessary visual complexity. Rather than consistently enhancing interaction, more detailed input may raise visual or temporal demands [27, 23];
- Consider how visual-feedback format influences attention and workload. During robot-assisted gait training, various visual representations resulted in different fixation and workload patterns [29];
- Maintain the visibility of relevant physical objects and avoid overlays that cause excessive occlusion, clutter, or tracking-related misalignment [24, 26]; and
- Avoid using visual feedback as the only channel for time-critical information when the primary task already necessitates sustained visual attention.

Therefore, visual feedback should be considered especially helpful for spatial precision and information richness, but its benefits must be balanced against the amount of visual attention available during the primary tasks.

D. Multimodal Combination Guidelines

The reviewed evidence does not support adding modalities simply because additional channels are available. Multimodal feedback is most beneficial when individual modalities provide complementary information or when redundancy is justified by the consequences of missing a critical cue [31, 32, 33].

Designers should:

- Assign a clear informational role to each modality instead of delivering the same information over every channel;
- Use complementary modalities when communicating several kinds of information at the same time. For example, haptic feedback communicates directional correction, whereas visual feedback may communicate detailed spatial information [31];
- Combine haptics with audio when users need non-visual semantic and spatial instruction. Mobile landmark scanning was made possible via audio-haptic guidance, which eliminated the need for constant screen-focused involvement [33];
- Consider the learning needs of new cross-modal mappings. Although SonoHaptics’ abstract audio-haptic mappings resulted in a comparatively high initial workload, it also improved target-selection time in cluttered scenes. [39];
- Evaluate potential trade-offs between speed and accuracy. Although it took longer than AR-only interaction, combined visual-haptic AR feedback decreased alignment inaccuracy and overshoot [31];
- Implement redundancy selectively when failure to notice a cue could have serious consequences;

- Reduce unnecessary modalities when additional channels increase workload without producing a corresponding performance benefit [34, 35].

The central principle is therefore purposeful multimodality: Additional channels should be added not just because the system can display them, but also because they offer helpful information or essential redundancy.

E. Timing and Synchronization Guidelines

The timing of the feedback should closely match the system event, user activity, or environmental situation that it represents. Both inside a single modality and across several synchronized channels, timing is important.

Immediate haptic feedback reduced sequential interaction time from 11.58 s to 8.56 s in the study by Xu et al. [16]. In contrast, response delays of approximately 2–5 s disrupted conversational flow in context-aware cooking interaction [20]. Cross-modal timing is also important: Sharma et al. found that 50% of drivers detected audio–haptic asynchrony outside an interval of approximately –75 ms to 110 ms [36].

Designers should:

- Provide feedback that is close enough to the relevant system event or user action to maintain a specific causal relationship;
- Minimize unnecessary delays in interactions that are time-sensitive or safety-relevant;
- Synchronize modalities that represent the same event;
- Avoid cross-modal timing differences that cause related cues to be perceived as separate events [36];
- Use intermediate progress feedback when a system operation justifiably needs more processing time. In multi-step in-car interactions, periodic progress feedback improved perceived processing speed and trust while reducing workload [68];
- Measure and report end-to-end feedback latency when evaluating multimodal systems, particularly in precision or high-stakes applications [30].

The analyzed studies do not support a single universal synchronization threshold for all applications. Therefore, the right time should be chosen for the specific modalities, task, and necessary reaction.

F. Context-Aware Modality Selection

The cross-domain results demonstrate that rather than a fixed hierarchy of auditory, haptic, and visual input, modality selection should be based on task needs and available user resources.

When users’ hands are occupied, cooking serves as an example of the importance of context-aware audio teaching [20]. According to walking research, auditory feedback can direct movement without the need for continuous visual inspection [22]. According to accessibility studies, when visual information is not available, haptic feedback can take over as the main channel [17, 14]. When the

visual channel is still suitable for the task, detailed augmented reality information can enhance spatial localization in precision and surgical interaction [26].

Designers should therefore consider:

- Which motor and sensory resources are already being used for the primary task;
- whether the information is sequential, persistent, spatial, directional, semantic, or time-critical;
- whether it is safe to redirect visual attention to an interface;
- whether visual clutter or noise from the environment may interfere with a particular modality;
- how quickly the user needs to understand and react to the feedback;
- whether each candidate modality can be reliably recognized by the user;
- whether an additional modality can communicate information that cannot be conveyed efficiently through the primary feedback channel.

The system should prioritize the modality or combination that communicates the required information while minimizing disruption to the user’s primary task.

G. Personalization Guidelines

The findings show that feedback perception and preference are not uniform across users. This is particularly valid for haptic feedback, where performance can be affected by body location, perceptual thresholds, and calibration.

Distal arm-level feedback similarly revealed location-dependent variations in user experience [53], and Wear+Touch exhibited notable changes between finger, wrist, and forearm feedback locations [38]. Significant variation in temporal perception for remapped tactile gait feedback was observed by Jouybari et al. [54]. Because stimulation perception varies with user and skin characteristics, electrotactile systems also need to be calibrated individually [13].

Designers should provide mechanisms to personalize, where appropriate:

- haptic intensity, duration, pattern, and stimulation location;
- auditory volume, sound mapping, and speech presentation;
- notification and feedback frequency;
- timing or sensitivity thresholds;
- the amount of visual information presented;
- the degree of multimodal redundancy;
- modality availability for users with sensory or physical impairments.

Personalization should not simply increase signal strength. Instead, feedback characteristics should be adapted to the user’s perceptual abilities, preferences, task performance, and interaction context.

H. Safety and Accessibility Guidelines

Safety and accessibility should be treated as core requirements when feedback is used during driving, walking, mobility assistance, clinical procedures, or other contexts in which missing or incorrect information could have serious consequences.

Instead of serving as vision supplements, auditory and haptic input may serve as the main information channels for blind and low-vision interaction. Visually impaired participants were able to locate spatial targets more effectively with shape-changing haptic guidance than with vibration alone, according to Quinn et al. [17]. Haptic feedback has been used for job tasks, training, navigation, and graphical comprehension, according to the wider accessibility evidence [14].

Timing and accuracy are especially important in safety-critical interactions. While AR guidance decreased localization error during laparoscopic target identification [26], audio-haptic misalignment increased annoyance and decreased perceptual coherence in driving [36]. Further evidence that enhanced usability does not always translate into increased physical safety comes from outdoor mobile interaction: the audio-haptic scanning system assessed by Van Brummelen et al. [33] improved a number of interaction outcomes but did not considerably lower the frequency of collisions.

Designers should:

- provide precise, unambiguous feedback on critical events;
- offer a non-visual alternative when obtaining essential information through visual means is unsafe or unreliable;
- use complementary redundancy in situations where missing a crucial cue could have serious consequences;
- differentiate urgent alerts from normal status information;
- make sure that additional feedback doesn't affect how environmental hazards are perceived;
- analyze cross-modal synchronization and system delay for time-sensitive interaction;
- Whenever possible, evaluate accessible feedback with the intended target users; and
- Evaluate safety directly rather than assuming that improvements in usability, accuracy, or preference necessarily translate into improved safety.

Overall, a task-dependent approach to multimodal feedback design is supported by these guidelines. There is no universal hierarchy that should be used to rank audio, tactile, and visual feedback. Rather, the information being communicated, the user's sensory and motor resources, and the demands of the primary task should all be taken into consideration when choosing the modality, encoding, timing, position, degree of redundancy, and degree of personalization.

VIII. RESEARCH GAPS AND FUTURE DIRECTIONS

Although the reviewed evidence indicates that auditory, haptic, visual, and multimodal feedback can support interaction under visual and manual constraints, it also reveals several persistent limitations in the current evidence base. These include short evaluation periods, limited ecological validity, individual variability, fixed feedback configurations, cognitive workload, synchronization requirements, heterogeneous evaluation methods, and the reliability of intelligent feedback systems. Addressing these limitations is necessary before findings from controlled evaluations can be generalized confidently to sustained real-world eyes-busy and hands-busy interaction.

A. Long-Term and Longitudinal Evaluation

The limited amount of longitudinal evaluation is a major limitation of the current evidence base. There is minimal information about learning, habituation, retention, or persistent acceptance because many of the studied systems were evaluated in single experimental sessions or brief interventions.

This limitation is present in a variety of feedback modalities. For example, [56] assessed vibrotactile postural feedback during brief simulated construction tasks and specifically noted the lack of long-term retention assessment as a limitation. Similarly, [64] assessed home-based visual feedback over roughly 5.8 weeks but did not perform long-term retention testing following the intervention. Similarly, [33] noted that their audio-haptic mobile navigation system was limited by unstudied long-term novelty effects.

Therefore, future research should look at whether improvements observed during initial use remain stable over repeated sessions. Task performance, workload, feedback recognition, preference, habituation, and ongoing system use should all be measured in a longitudinal assessment. This is especially important for wearable and always-available feedback systems since repeated exposure might change how users perceive or react to a signal.

B. Personalization and User Variability

The evidence that has been evaluated shows that the placement or arrangement of the feedback signal, as well as the user, can significantly affect how effective feedback is. This is especially evident in haptic interaction.

Terenti and Vatavu [38] found that whether vibrotactile feedback was given close to the index finger, wrist, or forearm had major effects on enjoyment and perceived efficiency. Additionally, they found that finger-level feedback was significantly preferred over more distal arm locations in their related evaluation of distal arm-level feedback [53]. According to [54], there is significant individual variation in temporal perception, with a 50% motor-awareness threshold of 378 ± 239 ms. According to studies on electrotactile feedback, regular individual calibration is required because skin conditions and other

user-specific factors might affect how stimulation is perceived [13].

These outcomes suggest that a single fixed configuration might not be suitable for every user. Future studies should look into the intensity, length, encoding, body location, and combinations of modalities of individualized feedback. To determine whether adapting feedback to each person's unique perceptual and performance characteristics provides measurable advantages, personalization should be experimentally examined against fixed configurations.

C. Adaptive and Context-Aware Feedback

The dependence on set feedback configurations despite changes in the user's job state is a second ongoing flaw. The findings show that context-sensitive feedback can enhance interaction, but adaptive strategies are still relatively unexplored. [20] showed the benefits of context-aware proactive voice assistance during cooking, where the system used task context to reduce interaction breakdowns and provide more appropriate spoken support. On the other hand, [68] specifically noted the absence of adjustable timing as a limitation and employed a fixed five-second interval for intermediate audio-visual progress reports. Static haptic mappings lacked real-time customisation, according to Salous et al. [7].

Therefore, future systems should look at input that varies based on the task state at hand, workload, surrounding factors, prior user reactions, and the accessibility of sensory resources. Modality, timing, information density, intensity, and degree of repetition can all be altered. To determine whether adaptation offers advantages beyond the additional system complexity, these systems should be compared with static alternatives.

D. AI-Driven Feedback and Reliability

Many systems using large language models or computationally generated feedback are also included in the reviewed evidence base. These investigations show new possibilities for variable task support, but they also highlight dependability issues that need more research.

Although the Mango Mango conversational cooking helper obtained good perceived trustworthiness using an LLM-based design, the results also show extra dialog and semantic interpretation problems that may raise cognitive pressure [42]. While citing issues with speech recognition, intent interpretation, background noise, and camera occlusion, Vid2Coach similarly demonstrates the potential of intelligent task support [40]. Salous et al. [7] found that while AI-generated feedback for haptic creation could develop accurate representations for some textures, output variability remained a limitation.

Therefore, future research should evaluate AI-driven multimodal input in terms of consistency, predictability, timeliness, mistake recovery, and the effects of inaccurate feedback in addition to usability and novelty. When

intelligent feedback is applied to jobs that are physically demanding or safety-related, these needs become especially crucial.

E. Cognitive Workload and Information Density

The Results show that additional feedback does not necessarily produce better interaction. Understanding the amount of information that can be provided before multimodal support turns into visual or cognitive interference is an ongoing research topic.

The Hybrid collision-feedback condition produced the lowest collision-time ratio in visual interaction, but it also significantly increased task-completion time, according to [27]. Although SonoHaptics offered efficient audio-haptic support for gaze-based selection, its novel mappings resulted in a comparatively high initial workload, and error rose from 5.7% in the absence of distractions to 23.4% in the presence of three distractions [39]. [35] observed increased subjective workload for the multimodal interface while reporting improvements in a number of attention, mental-model, trust, and lane-position metrics in autonomous driving. Additionally, [34] discovered that while extra visual and haptic feedback decreased hesitation, it did not affect final pitch-tuning accuracy or completion time.

Therefore, the interaction between data density, redundancy, modality combinations, and primary-task workload should be investigated in future research. Rather than assuming that an extra sensory channel always enhances performance, multimodality should be evaluated in terms of both its advantages and its increased attentional costs.

F. End-to-End Latency and Synchronization

Another significant area of unfulfilled research needs is temporal coordination. Differences in time can lead to signals being interpreted independently rather than as a single cohesive feedback event when many modalities express the same event.

50% of participants identified audio-haptic irregularities outside of a window of roughly -75 ms to 110 ms in a driving situation, according to [36]. Tasaka [37] noted a decline in the user's sense of agency as buffering delay rose and found clear delay-related alterations in haptic—audio-visual communication. Cross-modal delay or sensory misalignment were found to be potential causes of decreased performance and involvement in multimodal teleoperation, which also needed extremely low processing latency [30].

Future research should examine acceptable synchronization ranges for particular workloads and modality combinations and report end-to-end system latency more consistently. A single universal temporal threshold for audio-haptic, visual-haptic, and audio-visual interaction is not supported by the available data.

G. Real-World and Ecologically Valid Evaluation

Many studies carried out in controlled labs, virtual reality settings, simulators, or simplified clinical settings are included in the updated results. Although these evaluations provide accurate measurements, they might not accurately capture the environmental complexity of actual hands-and-eyes-busy engagement.

Instead of utilizing real-world walking, [17] assessed non-visual haptic navigation using virtual targets in a controlled indoor environment. In a visually minimal virtual reality setting, [22] investigated auditory redirected walking. Using a stiff phantom that failed to replicate the tissue deformation experienced during actual surgery, [26] assessed surgical AR. Additionally, the driving assessments by [75] and [35] were not carried out in unrestricted road environments, but rather were simulator-based.

Because their audio-haptic navigation system was tested during an outdoor landmark-scanning job, [33] offer a significant exception. However, that study failed to identify long-term impacts and was limited to a single landmark.

Promising laboratory results should therefore be applied to real-world workplaces, kitchens, mobility environments, rehabilitation settings, and operational high-stakes activities in future studies. These assessments should look at whether feedback is still recognizable and helpful in the face of interruptions, movement, noise, visual clutter, changing lighting, and conflicting attentional demands.

H. Standardized Evaluation and Reporting

The outcome measures used in the examined studies are extremely diverse. Movement-error duration [21], trust and workload effect sizes [68], surgical localization error and NASA-TLX measures [26], haptic target localization performance [17], recognition accuracy and interaction time [16], and usability measures across multiple systems are some examples of the evidence.

Although it makes quantitative comparison challenging, this diversity mirrors the range of application domains addressed by the review. When appropriate, future studies should report a common core of metrics, such as job completion time, accuracy or error, response time, workload, usability, preference, safety-related consequences, and system latency. Additionally, studies should include sufficient data regarding participant characteristics, task conditions, encoding, synchronization, location, intensity, and length of feedback.

I. Accessibility and Target-Population Validation

Several studies also demonstrate a gap between intended users and evaluated participants. This is particularly important in accessibility and rehabilitation research.

Instead of using users with organic visual impairment, Ricci et al. [18] assessed multimodal navigation with healthy subjects undergoing simulated glaucoma. Instead

of using stroke patients with motor impairments, Luciani et al. [24] used healthy individuals to assess an AR-based rehabilitation system. On the other hand, [17] compared shape-changing and vibrotactile navigation with individuals who were visually impaired.

Therefore, future evaluations should include the intended target population whenever feasible. Findings obtained from healthy participants or simulated impairments should not automatically be assumed to generalize to users with sensory, motor, or cognitive impairments.

J. Privacy, Transparency, and User Control

In the reviewed data, privacy was evaluated less frequently than usability or performance, although it becomes more crucial for context-aware systems. For example, Jaber et al. found privacy issues with gathering continuous work context for proactive voice interaction [20].

Future adaptive systems may require information about voice, movement, physiological state, gaze, environmental context, or interaction history. Research should therefore examine how such data are collected, processed, stored, and communicated to users, particularly when they are used to adapt feedback automatically. Privacy, transparency, and user control should be treated as evaluation requirements rather than solely as implementation concerns.

A major research opportunity is to move from short-term, fixed-feedback demonstrations toward systems evaluated over longer periods, under real-world conditions, and with adaptation to task and user context. However, adaptive and AI-supported approaches should be compared experimentally with simpler fixed-feedback configurations rather than being assumed to provide superior outcomes.

K. Conceptual Framework for Adaptive Multimodal Feedback

Based on the recurring findings and research gaps identified in this review, we propose a conceptual framework for future adaptive multimodal feedback systems. The framework is not presented as a validated system architecture; rather, it organizes the synthesized evidence into a closed-loop model intended to guide future system design and evaluation.

The framework contains five interconnected stages: *Context Sensing*, *Feedback Decision*, *Multimodal Feedback*, *User Response*, and *Continuous Adaptation*.

Context Sensing: The first stage involves identifying information relevant to the current interaction context. Task status, user activity, accessible sensory resources, environmental variables, and interaction history may all be included, depending on the application. Context-aware cooking engagement, where task-state information was used to provide more appropriate proactive voice assistance, demonstrates the significance of task context [20]. Therefore, rather than assuming that the same modality

is suitable throughout an encounter, context should be considered as an input to feedback selection.

Feedback Decision: The second step establishes the information that needs to be shared and the most suitable media. According to the cross-domain synthesis, visual feedback is suitable for detailed or persistent information when visual attention is available, haptic feedback is particularly suitable for localized, directional, spatial, or immediate information, and auditory feedback is suitable for semantic and sequential guidance [16, 19, 17, 26]. Determining whether different channels offer truly complementary information or just duplicate one another should also be part of the decision-making process.

Multimodal Feedback: After that, the selected information is communicated through auditory, visual, haptic, or multimodal feedback. Multimodal combinations can enhance certain results when the channels offer complementary support, according to evidence from AR guidance, VR interaction, mobility assistance, and mobile scanning [31, 32, 18, 33]. However, additional modalities should not be activated by default because multimodal feedback can also introduce additional workload or task time [34, 35].

User Response: The system evaluates how well the user responds to the feedback presentation. Relevant observations could include response time, task accuracy, errors, workload, feedback recognition, usability, or interaction success, depending on the task. The reviewed studies show that various outcomes, such as interaction time [16], movement error [21], spatial accuracy [26], and workload or trust [68], can reflect the effectiveness of feedback.

Continuous Adaptation: The last step makes decisions about feedback based on observed interaction outcomes. When the existing configuration results in poor performance or an excessive burden, adaptation may entail altering modality, intensity, encoding, body position, timing, or information density. Evidence that feedback effectiveness varies with body location [38], temporal alignment [36], and task context [20] supports the need for such adaptation. However, rather than being a mechanism confirmed by the current analysis, the adaptive loop is still a suggestion for more research.

The conceptual framework can therefore be represented as:

**Context Sensing → Feedback Decision →
Multimodal Feedback → User Response →
Continuous Adaptation → Context Sensing**

Certain features of this system, such as context interpretation, feedback creation, and choosing between several feedback configurations, may be supported by artificial intelligence and machine learning. Both this commitment and its limitations are demonstrated by current intelligent task-assistance research. For instance, AI-created haptic representations displayed variability across outputs [7], whereas LLM-based cooking interaction supported flex-

ible conversation but also generated semantic interpretation mistakes [42]. Therefore, rather than being proof that adaptive feedback is intrinsically better, AI should be viewed as an enabling implementation tool rather than as evidence that adaptive feedback is inherently superior.

In order to determine whether adaptation results in measurable advances in performance, workload, usability, safety, and user experience, future assessments of this approach should compare adaptive feedback with fixed configurations. Long-term behavior, dependability, privacy, and user control should all be considered in this assessment.

These research directions are not equally mature. Some, such as synchronization, workload, and feedback location, are supported by direct empirical evidence showing measurable differences between feedback configurations. In contrast, fully adaptive and AI-driven multimodal feedback remains an emerging direction. Future work should therefore validate adaptive approaches against fixed baselines rather than assuming that increased system complexity will necessarily improve interaction.

IX. LIMITATIONS AND CHALLENGES

Despite the review's contributions, it is important to recognize a few limitations while analyzing its results.

- 1) **Dependency on original study quality:** This analysis is fundamentally dependent on the quality, methodological rigor, and reporting standards of the included studies because it is a systematic literature review. The evidence compiled in this review may have been impacted by variations in trial design, participant selection, feedback implementation, and reporting procedures. Therefore, the conclusions derived from the total body of information may potentially be impacted by limitations or biases in the original studies.
- 2) **Search scope and time frame:** Within a predetermined publication period, the evaluation was carried out using a predetermined collection of educational databases and search phrases. Even if a number of important sources were taken into account, relevant research that were published in different settings or indexed outside of the chosen databases might not have been found. In the same way, significant technical reports, theses, gray literature, and other unpublished material might have been overlooked by the search technique. These limitations may have decreased the completeness of the recovered material, but they were required to keep the review scope narrow and manageable.
- 3) **Heterogeneity of studies and evaluation methods:** The application domain, feedback mechanism, task features, participant populations, experimental settings, and evaluation measures varied significantly among the included studies. For instance, various performance metrics and experimental techniques

TABLE X: Research Gaps and Corresponding Research Directions

Research Gap	Evidence from the Review	Recommended Research Direction
Long-term and real-world evaluation	Many studies used short laboratory, VR, simulator, or limited-session evaluations, with long-term retention or novelty effects often untested [56, 64, 33, 26].	Conduct longitudinal and field studies to evaluate sustained performance, habituation, acceptance, and robustness under realistic movement, noise, interruptions, and environmental conditions.
Personalization and target-user validation	Feedback effectiveness varied with body location and individual perception, while some assistive and rehabilitation systems were tested with healthy or proxy participants [38, 54, 18, 24].	Evaluate adjustable intensity, timing, encoding, location, and modality with diverse users and intended target populations rather than assuming one configuration generalizes to everyone.
Adaptive and AI-driven feedback	Context-aware systems showed potential, but fixed configurations and AI-based systems still exhibited semantic errors, output variability, and limited adaptation [20, 68, 42, 7].	Develop context-aware adaptation of modality, timing, and information density, and compare adaptive or AI-driven systems directly with fixed baselines for reliability and performance.
Cognitive workload and information density	Additional modalities did not always improve performance and could increase workload, clutter, or task time [39, 34, 35].	Determine when complementary multimodality improves performance and when additional cues become unnecessary perceptual or cognitive overhead.
Synchronization and latency	Cross-modal timing affected perception and agency; audio-haptic asynchrony was detectable outside approximately -75 to 110 ms [36, 37, 30].	Report end-to-end latency consistently and establish task- and modality-specific synchronization requirements.
Standardized evaluation and reporting	Studies used heterogeneous measures of accuracy, time, workload, usability, trust, movement performance, and user experience [16, 21, 26].	Adopt a common core of performance, error, response-time, workload, usability, safety, and latency measures, together with clear reporting of feedback parameters and participant characteristics.

were frequently employed in research, including driving, cooking, physical activity, accessibility, walking, and high-stakes tasks. This diversity restricts the degree to which a single overall success measure can be produced and makes direct comparison between trials challenging. As a result, rather than a clear numeric ranking of modalities, the synthesis mainly focuses on repetitive evidence patterns.

- 4) Sample size and population characteristics: Many studies in the literature review relied on particular user communities or comparatively small participant samples. The diversity of users who potentially engage with multimodal feedback systems in real-world contexts may not be properly represented by some tests, which were carried out with youthful or technically proficient participants. Feedback evaluation and usage may vary depending on age, physical capabilities, sensory capacities, technological expertise, and task familiarity. As a result, some findings' applicability to larger populations is still restricted.
- 5) Short-term and controlled evaluations: A large number the systems under consideration were evaluated using controlled laboratory tasks or brief experimental sessions. Long-term usability, learning effects, habituation, sustained workload, and ongoing user approval are not well covered by such research, despite the fact that they offer helpful evidence regarding instant performance. As a result, the effectiveness seen during the first encounter might not be the same as the effectiveness following repeated or extended use. To determine whether

observed benefits hold over time, further long-term and ecologically valid evaluations are needed.

- 6) Differences in feedback implementation: Studies might use the same modality in quite diverse ways. For example, while audio feedback may vary in speech content, volume, timing, and sound design, haptic feedback may vary in intensity, frequency, duration, placement, and encoding. It is challenging to relate observed performance variations to the modality alone due to these implementation differences. Therefore, the results of this review should be evaluated in light of the particular feedback implementations and task contexts that the original studies looked into.
- 7) Uneven evidence across domains: The amount and directness of the evidence vary depending on the application setting. While some domains have fewer direct real-world studies, others have multiple controlled evaluations. Therefore, rather than demonstrating that a single feedback configuration will function consistently across driving, cooking, rehabilitation, accessibility, mobile interaction, and high-stakes tasks, the cross-domain recommendations should be read as recurring evidence patterns.
- 8) Limited quantitative comparability: The reported metrics and experimental conditions were not sufficiently consistent to support a pooled statistical comparison. Although many publications reported quantitative changes in outcomes such as accuracy, task-completion time, response time, error, or workload, differences in study design, measurement methods, participant populations, and reporting formats prevented a valid meta-analysis. Consequently,

the numerical results presented in this review should be interpreted as findings from individual studies rather than as directly comparable estimates of overall modality effectiveness.

- 9) Rapid technological development: AI-based interfaces, wearable technology, augmented reality, intelligent systems, and multimodal interaction technologies are all developing quickly. As a result, as new sensing technologies, adaptive algorithms, and feedback mechanisms are developed, the technological approaches and conclusions mentioned in this analysis may become less representative. Therefore, capabilities that were not properly reflected in the current body of evidence may be introduced by future research.
- 10) Study-level quality assessment: No formal study-level risk-of-bias instrument was applied. The Strong/Moderate/Limited classifications used in this review therefore represent synthesis-level evidence judgments based on consistency, directness, and the number and limitations of supporting studies rather than standardized methodological-quality ratings.

Despite these limitations, the review provides a structured synthesis of research on multimodal feedback for eyes-busy and hands-busy interaction. The identified limitations highlight the need for larger and more diverse participant samples, longitudinal and real-world evaluation, standardized outcome measures, validation with intended target populations, and more consistent reporting of feedback characteristics.

X. CONCLUSION

This systematic literature review synthesized evidence on auditory, haptic, visual, and multimodal system-to-user feedback in interactions where visual attention, manual resources, or both are constrained by a primary task. The reviewed evidence covered driving, cooking and domestic tasks, physical activity and rehabilitation, accessibility, walking and mobile interaction, extended reality, and surgery or other high-stakes precision tasks.

Across these contexts, no single feedback modality was universally superior. Haptic feedback was most consistently associated with immediate, localized, directional, and spatial information. Auditory feedback was particularly suitable for semantic, sequential, explanatory, and movement-related guidance, while visual feedback remained valuable for detailed, persistent, and spatially precise information when sufficient visual attention was available.

Multimodal feedback produced benefits when individual channels served clear and complementary roles rather than simply duplicating information. However, additional modalities could also introduce greater workload, longer task times, learning requirements, or synchronization costs. Purposeful multimodality is therefore more

strongly supported than simply adding more feedback channels.

Beyond modality selection, the review identifies timing, synchronization, body location, signal encoding, information density, cognitive workload, task context, accessibility, and user variability as important design factors. Future research should emphasize longitudinal and real-world evaluation, validation with intended target populations, personalization, context-aware adaptation, end-to-end latency reporting, reliable AI-generated feedback, and standardized evaluation measures.

Overall, eyes-busy and hands-busy interaction should be treated as a context-dependent feedback-design problem. Effective systems should provide the required information through an appropriate modality at an appropriate time while minimizing interference with the user's primary task.

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